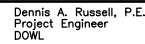
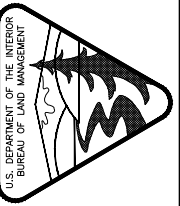
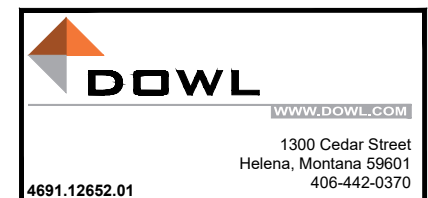




MALTA FIELD OFFICE



SHEET	DRAWING	DESCRIPTION	CAD DRAWING
1	G1	COVER SHEET AND SHEET INDEX	SC-GN-CV-12652-01.dwg
2	G2	LOCATION AND VICINITY MAPS	SA-CS-LV-12652-01.dwg
3	G3	GENERAL STRUCTURAL NOTES & DETAILS	SC-GN-ST-12652-01.dwg
4	G4	RATING CURVES	SC-GN-ST-12652-01.dwg
5	C1	SITE PLAN AND SURVEY CONTROL	SA-CS-EX-12652-01.dwg
6	C2	DAM EMBANKMENT EXCAVATION PLAN AND PROFILE	SC-CS-EMB-EXCV-12652-01.dwg
7	C3	DAM EMBANKMENT PLAN AND PROFILE	SC-CS-EMB-CREST-FG-12652-01.dwg
8	D1	TOE DRAIN DETAILS	SC-CS-EMB-CREST-FG-12652-01.dwg
9	EX1	DAM EMBANKMENT SECTIONS STA. 3+50 TO STA. 4+25	SC-CS-EMB-CREST-XS-12652-01.dwg
10	EX2	DAM EMBANKMENT SECTIONS STA. 4+50 TO STA. 5+00	SC-CS-EMB-CREST-XS-12652-01.dwg
11	EX3	DAM EMBANKMENT SECTIONS STA. 5+25 TO STA. 5+75	SC-CS-EMB-CREST-XS-12652-01.dwg
12	EX4	DAM EMBANKMENT SECTIONS STA. 6+00 TO STA. 6+50	SC-CS-EMB-CREST-XS-12652-01.dwg
13	EX5	DAM EMBANKMENT SECTIONS STA. 6+75 TO STA. 7+25	SC-CS-EMB-CREST-XS-12652-01.dwg
14	OW1	OUTLET WORKS DEMOLITION AND EXCAVATION PLAN	SC-CS-OW-DT-12652-01.dwg
15	OW2	OUTLET WORKS PLAN AND PROFILE	SC-CS-OW-DT-12652-01.dwg
16	OW3	OUTLET WORKS TYPICAL SECTIONS AND DETAILS	SC-CS-OW-DT-12652-01.dwg
17	OW4	OUTLET WORKS TOWER DETAILS	SC-CS-OW-DT-12652-01.dwg
18	OX1	OUTLET WORKS SECTIONS STA. 0+75 TO STA. 1+50	SC-CS-OW-DT-12652-01.dwg
19	OX2	OUTLET WORKS SECTIONS STA. 1+75 TO STA. 2+25	SC-CS-OW-DT-12652-01.dwg
20	OX3	OUTLET WORKS SECTIONS STA. 2+50 TO STA. 3+75	SC-CS-OW-DT-12652-01.dwg
21	SB1	STILLING BASIN PLAN AND PROFILE	SC-CS-OW-DT-12652-01.dwg
22	SW1	SPILLWAY IMPROVEMENTS PLAN AND PROFILE	SC-CS-ASW-12652-01.dwg
23	SW2	SPILLWAY IMPROVEMENTS PLAN, PROFILE, & DETAILS	SC-CS-ASW-12652-01.dwg
24	SWX1	SPILLWAY CROSS SECTIONS STA. 0+00 TO STA. 3+25	SC-CS-ASW-12652-01.dwg
25	SWX2	SPILLWAY CROSS SECTIONS STA. 3+50 TO STA. 6+75	SC-CS-ASW-12652-01.dwg
26	SWX3	SPILLWAY CROSS SECTIONS STA. 7+00 TO STA. 7+75	SC-CS-ASW-12652-01.dwg



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BUREAU OF LAND MANAGEMENT

COVER SHEET AND SHEET INDEX

BAR ISLAND DAM MODIFICATIONS

MONTANA

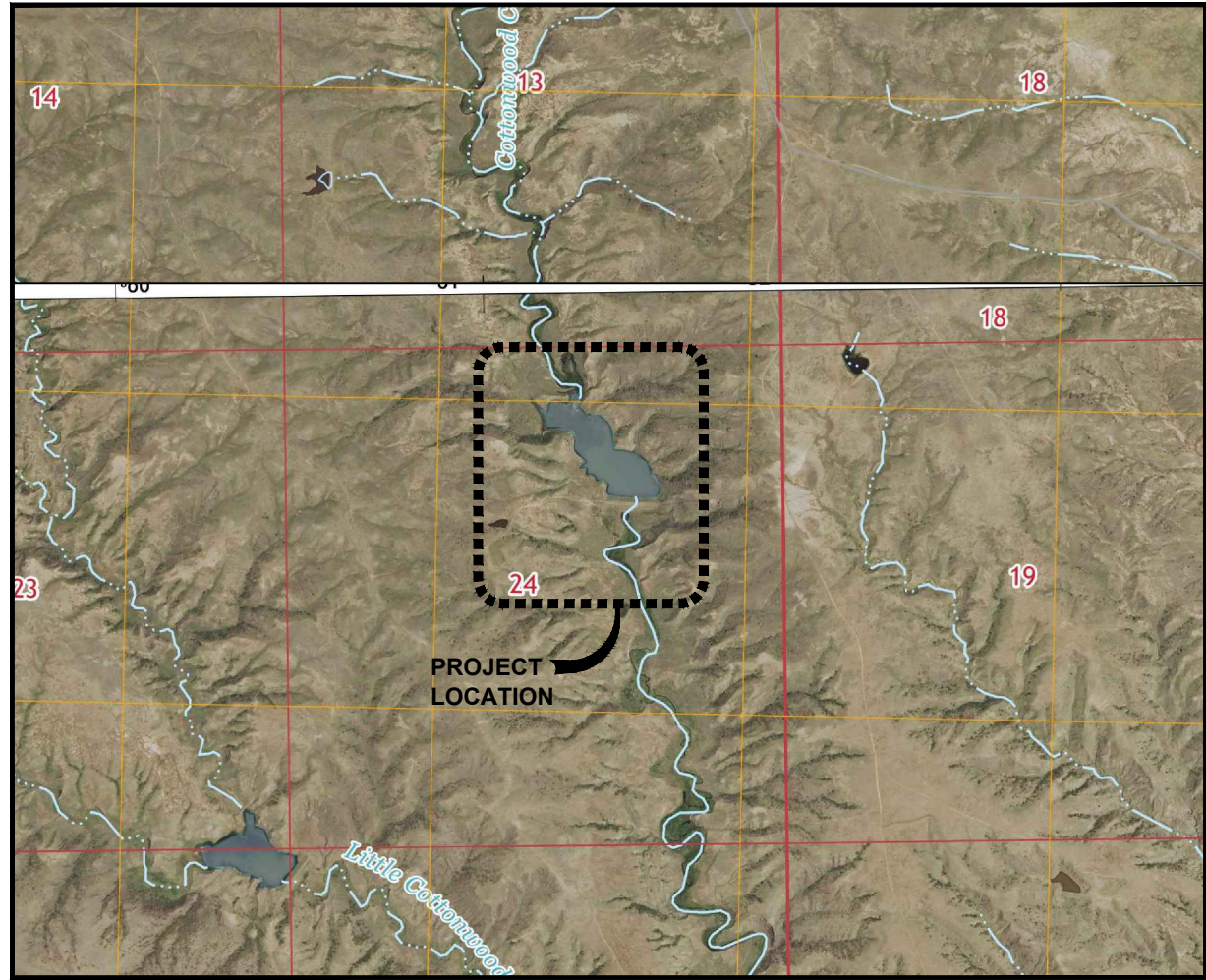
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G1

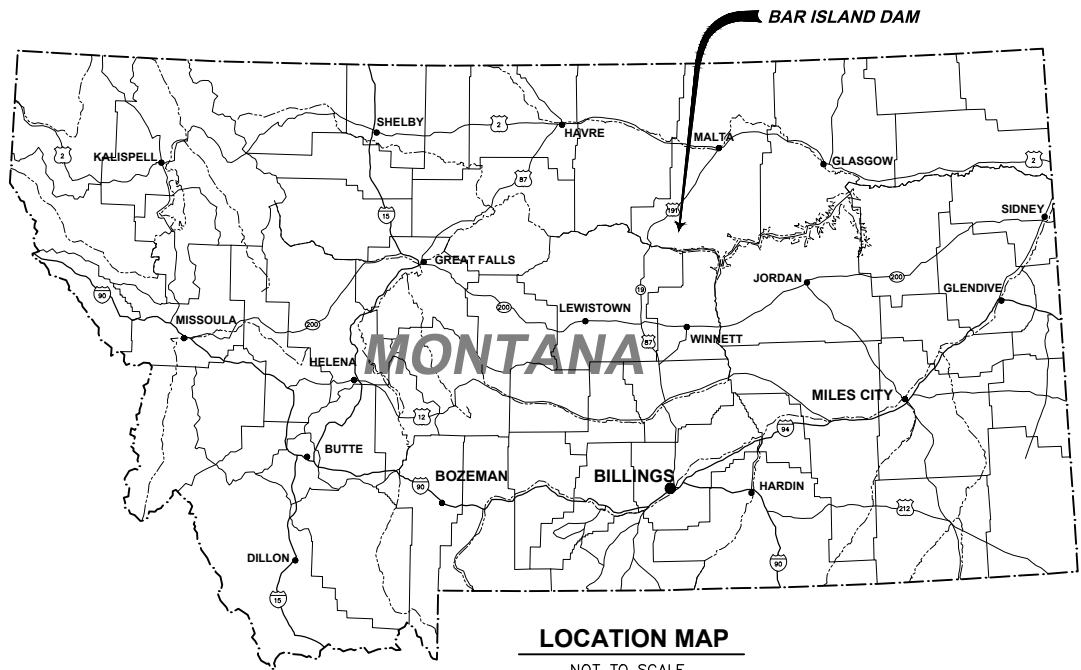
SHEET 1 OF 26

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INSET MAP

1000 0 1000 2000
SCALE IN FEET

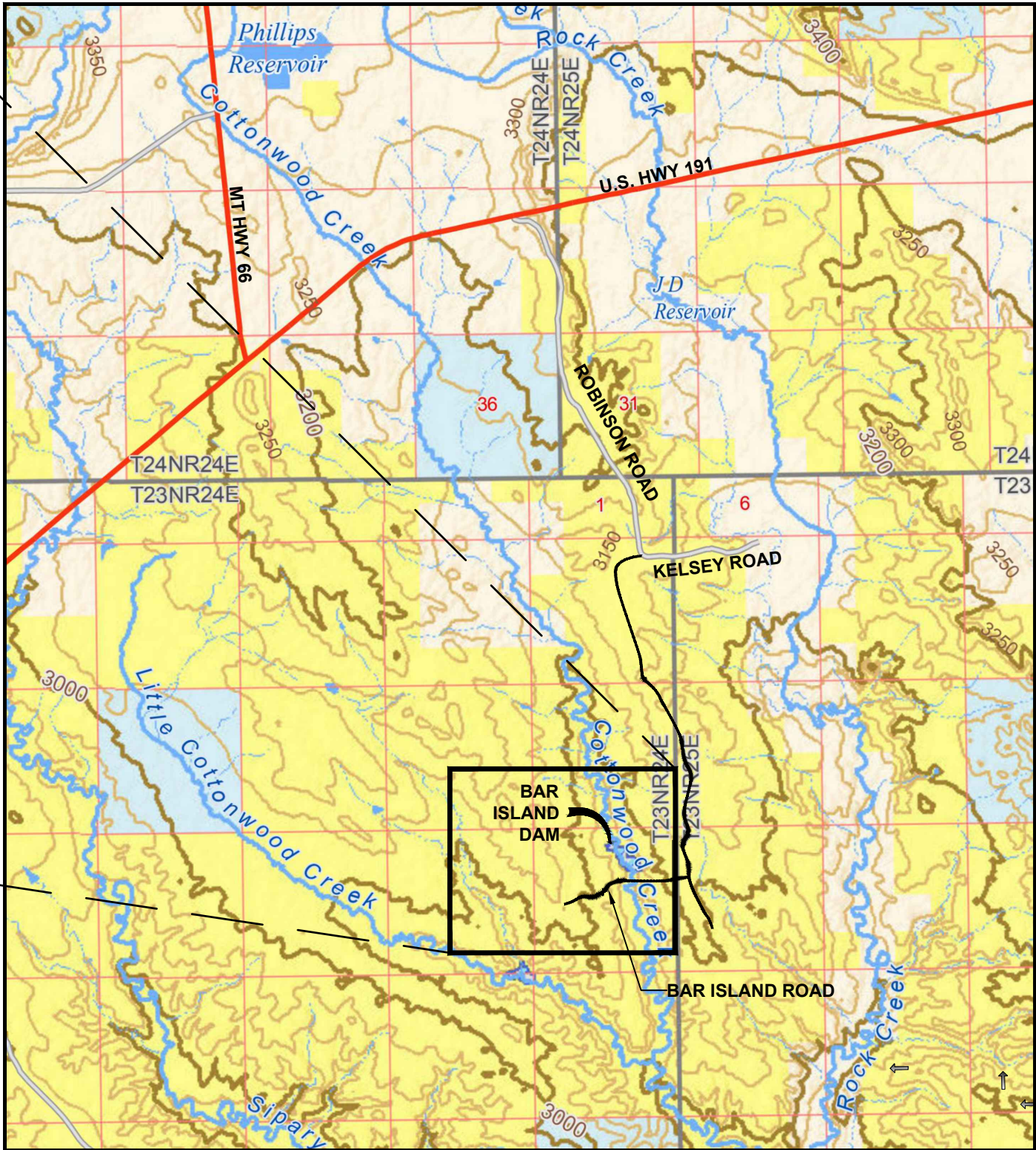


LOCATION MAP

NOT TO SCALE

Driving Directions:

TRAVEL SOUTH FROM MALTA ON US HIGHWAY 191 S FOR APPROXIMATELY 53 MILES. TURN LEFT ONTO ROBINSON ROAD AND TRAVEL 2.7 MILES. TURN RIGHT ON BAR ISLAND ROAD AND TRAVEL 2.5 MILES. TURN RIGHT ON TWO TRACK AND TRAVEL 0.3 MILES TO BAR ISLAND RESERVOIR.



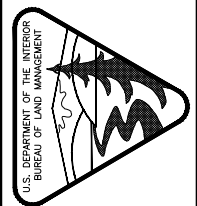
LAND OWNERSHIP

-  BLM
-  STATE
-  PRIVATE

VICINITY MAP

2500 0 2500 5000
SCALE IN FEET





UNITED STATES DEPARTMENT OF THE INTERIOR
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LOCATION AND VICINITY MAPS

BAR ISLAND DAM MODIFICATIONS

MALTA MONTANA

AUTOCAD NAME:	SA-CS-LV-12652-01.DWG
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)
DESIGNED BY:	DAR
DRAWN BY:	IME
CHECKED BY:	RGR
APPROVED BY:	DAR
DATE:	12-22-25
DATE:	12-22-25
DATE:	12-22-25
DATE:	12-22-25

G2

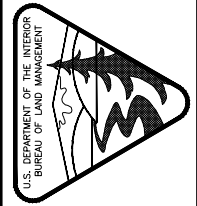
SHEET 2 OF 26

- 1. APPLICATION OF SPECIFICATIONS AND CODES
 - a. CONSTRUCTION AND DESIGN SHALL BE IN ACCORDANCE WITH THE AMERICAN CONCRETE INSTITUTE SPECIFICATIONS AND THE REQUIREMENTS NOTED AS FOLLOWS
- 2. SPECIAL INSPECTION
 - a. CONCRETE -- TABLE 1705.3 OF THE IBC
 - i. PERIODIC INSPECTION OF REINFORCEMENT BEFORE CONCRETE IS PLACED
 - ii. FULL-TIME INSPECTION OF CAST IN AND POST-INSTALLED ANCHOR RODS AS IDENTIFIED HEREIN
 - iii. FULL-TIME INSPECTION DURING PLACEMENT OF CONCRETE, INCLUDING THE TAKING OF TEST SPECIMENS, SLUMP, AND AIR, BE LIMITED TO STRUCTURAL REINFORCED CONCRETE WITH TESTING FREQUENCY IN ACCORDANCE WITH THE PROJECT TECHNICAL SPECIFICATIONS
 - b. STEEL -- AISC 360 FOR STRUCTURAL STEEL, IBC TABLE 1705.2.3 FOR STEEL CONSTRUCTION OTHER THAN STRUCTURAL STEEL (COLD-FORMED STEEL, REBAR, ETC.)
- 3. DESIGN LOADS
 - a. DESIGN LOADS AND LOAD APPLICATIONS SHALL BE IN ACCORDANCE WITH IBC
 - b. UNIFORM DEAD AND LIVE LOADS
 - i. DEAD LOADS, UNLESS OTHERWISE NOTED, SELF-WEIGHT OF STRUCTURE AND APPURTENANCES
- 4. CONSTRUCTION LOADS
 - a. STRUCTURES HAVE BEEN DESIGNED FOR DEAD LOADS, AND THE DESIGN LOADS NOTED ABOVE. PROVIDE TEMPORARY BRACING, SHORING, OR OTHER SUPPLEMENTAL SUPPORT DURING CONSTRUCTION AS NECESSARY TO PROTECT THE STRUCTURES FROM EXCESSIVE CONSTRUCTION LOADS
- 5. FOUNDATIONS
 - a. BACKFILL ON WALLS WITH FILL ON BOTH SIDES SHALL BE COMPACTED IN EQUAL LIFTS EACH SIDE OF WALL. WALLS BACKFILLED FROM ONE SIDE ONLY SHALL HAVE ALL SUPPORTING SLABS, PERMANENT FRAMING, OR TEMPORARY BRACING IN PLACE BEFORE PLACEMENT OF BACKFILL.



NOTE:
1. STANDARD LAP; SEE GENERAL NOTE 6G

BAR SIZE	a	b
#4	18"	24"
#5	24"	30"
#6	30"	36"
#7	42"	54"
#8	48"	62"
#9	54"	70"
#10	61"	79"



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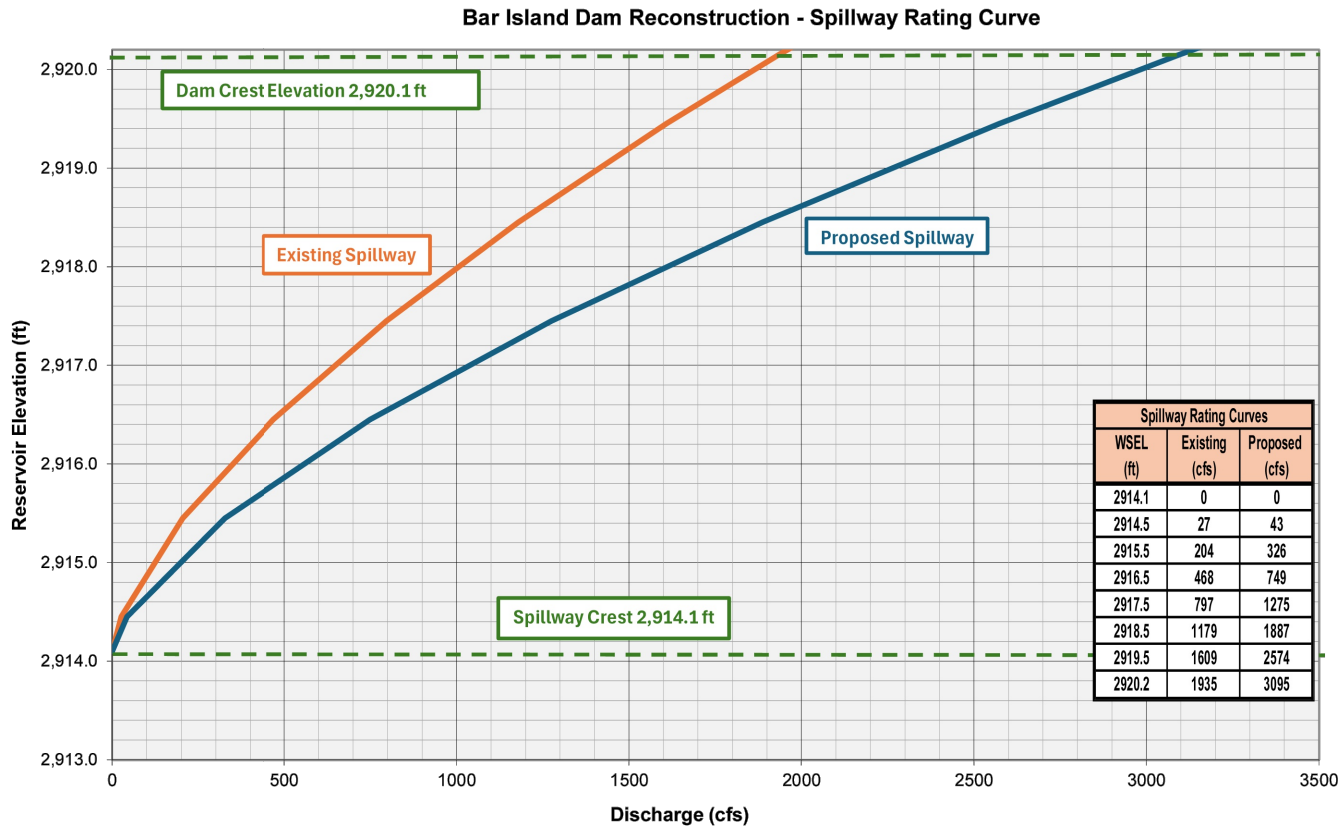
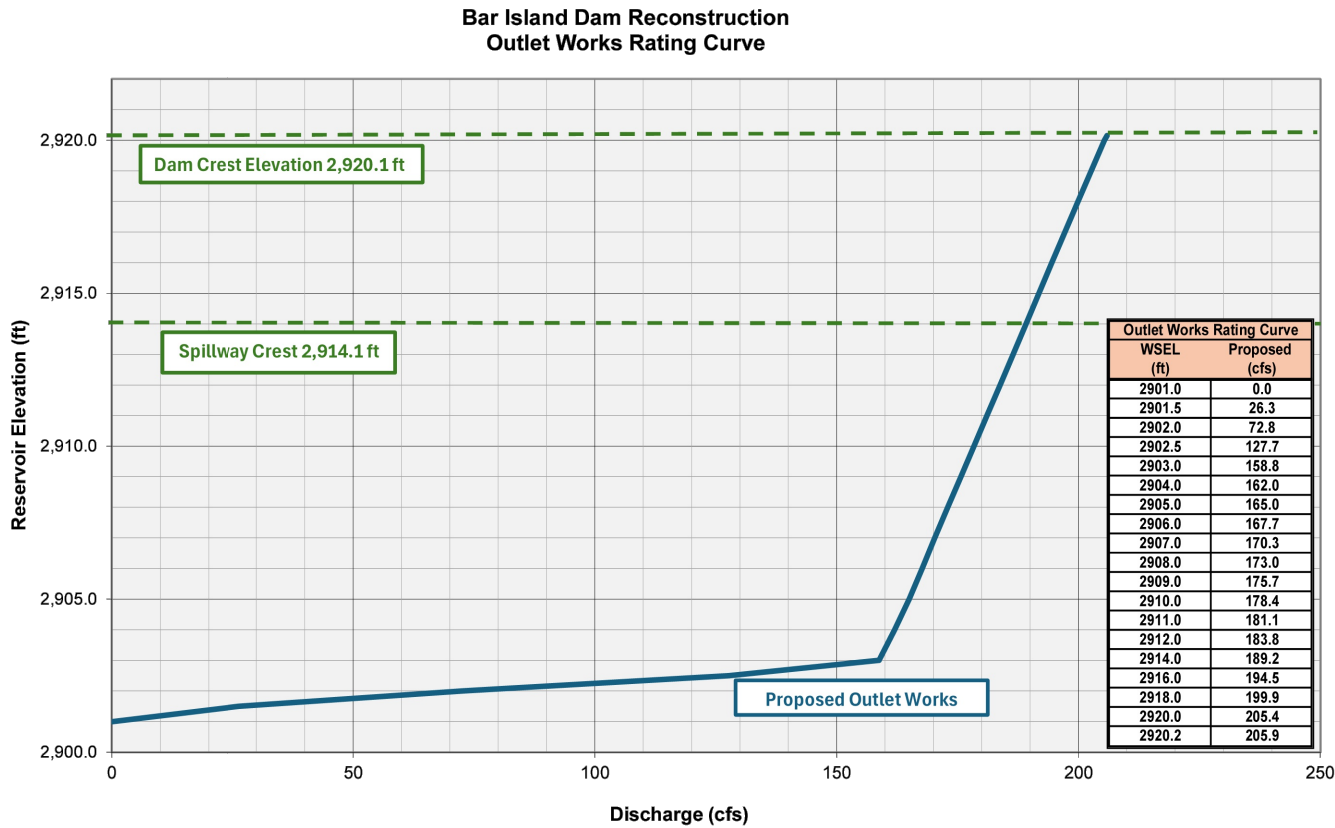
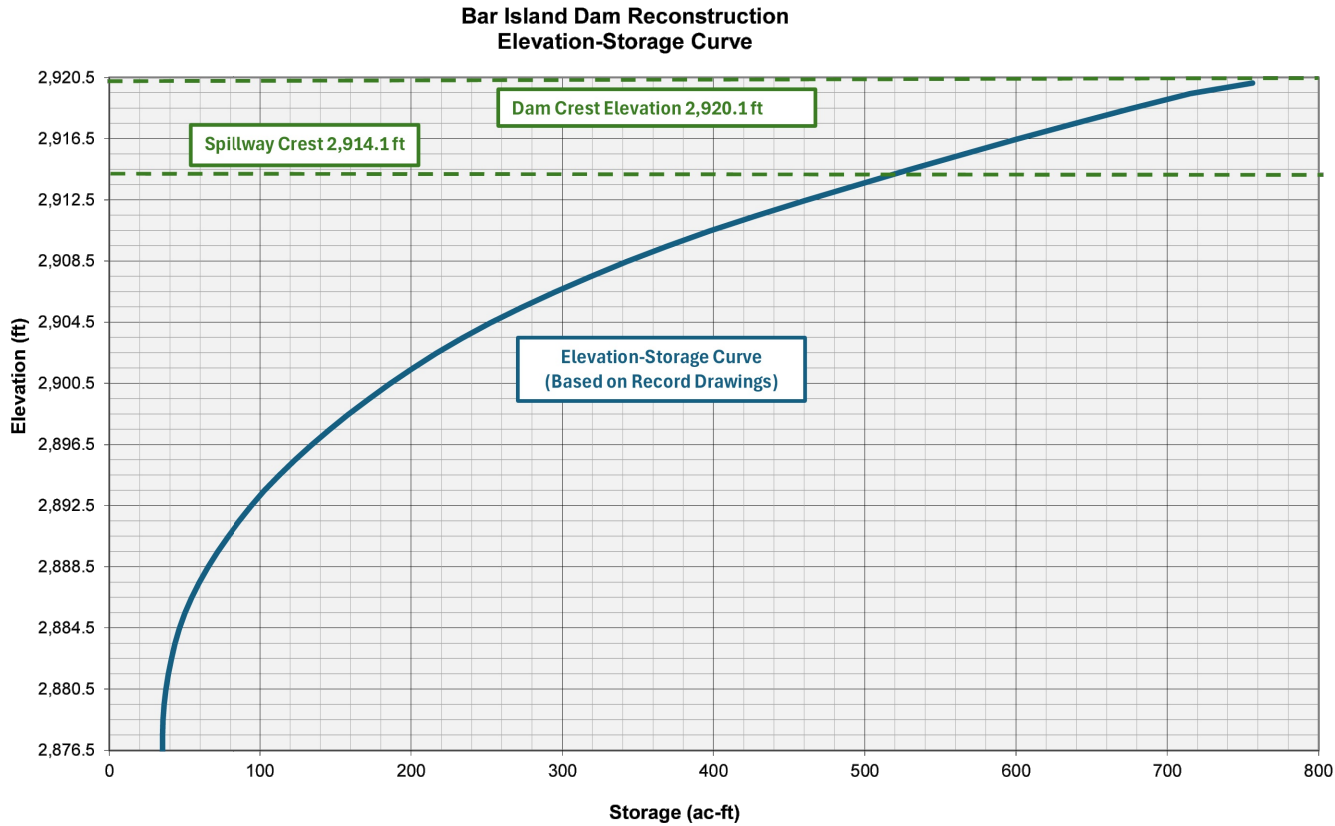
**GENERAL STRUCTURAL
NOTES & DETAILS**

BAR ISLAND DAM MODIFICATIONS

AUTOCAD NAME:	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY: DAR	DATE: 12-22-25			
DRAWN BY: IME	DATE: 12-22-25			
CHECKED BY: RCR	DATE: 12-22-25			
APPROVED BY: DAR	DATE: 12-22-25			

G3
SHEET 3 OF 26

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RATING CURVES

BAR ISLAND DAM MODIFICATIONS

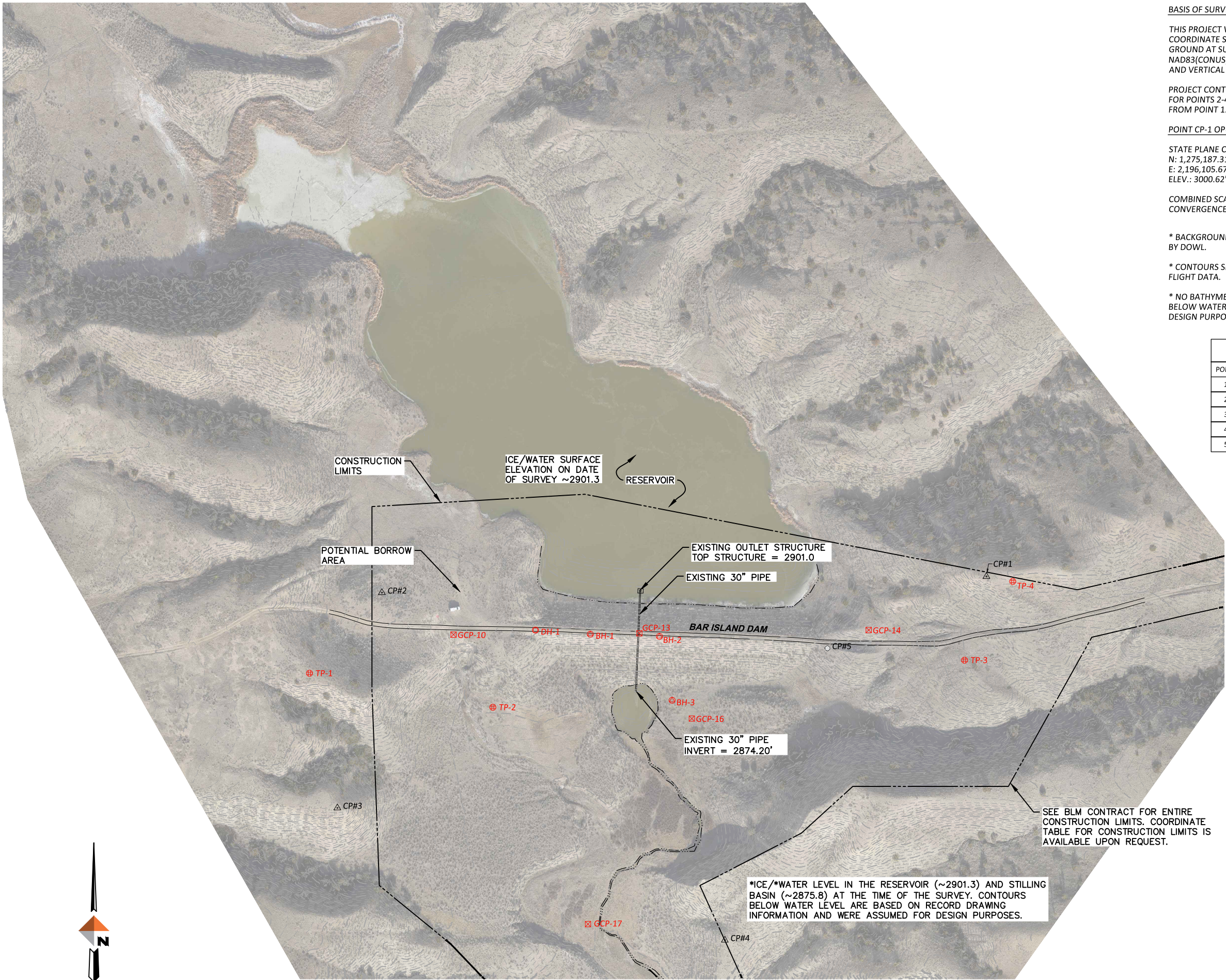
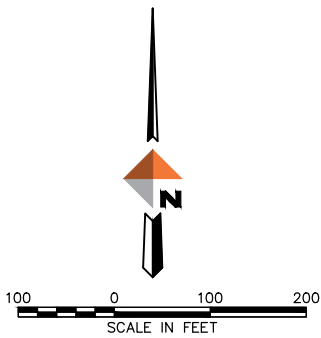
MONTANA

AUTOCAD NAME:	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY: DAR			DATE: 12-22-25	
DRAWN BY: IME			DATE: 12-22-25	
CHECKED BY: RCR			DATE: 12-22-25	
APPROVED BY: DAR			DATE: 12-22-25	

G4

SHEET 4 OF 26

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BASIS OF SURVEY

THIS PROJECT WAS SURVEYED ON THE MONTANA ZONE 2500 STATE PLANE COORDINATE SYSTEM USING RTK GPS. THE PROJECT WAS BROUGHT TO GROUND AT SURVEY CONTROL POINT CP-1. THE GEODETIC DATUM IS NAD83(CONUS). THE VERTICAL DATUM IS NAVD88 (GEOID18). HORIZONTAL AND VERTICAL UNITS ARE SURVEY FEET AT GROUND DISTANCES.

PROJECT CONTROL WAS ESTABLISHED BY OPUS POSITION AT POINT 1. VALUES FOR POINTS 2-4 WERE DETERMINED BY REDUNDANT RTK GPS OBSERVATIONS FROM POINT 1.

POINT CP-1 OPUS POSITION:

STATE PLANE COORDINATES:
N: 1,275,187.31'
E: 2,196,105.67'
ELEV.: 3000.62'

COMBINED SCALE FACTOR: 0.99933335
CONVERGENCE: 0.67705278"

* BACKGROUND IMAGE IS FROM DECEMBER 2024 DRONE FLIGHT COMPLETED BY DOWL.

* CONTOURS SHOWN ARE GENERATED FROM THE DECEMBER 2024 DRONE FLIGHT DATA.

* NO BATHYMETRIC SURVEY WAS PERFORMED. CONTOURS WERE PROJECTED BELOW WATER LEVEL BASED ON RECORD DRAWING INFORMATION FOR DESIGN PURPOSES.

POINT COORDINATE TABLE				
POINT	NORTHING	EASTING	ELEVATION	DESCRIPTION
1	1275187.31	2196105.67	3000.617	2" A.C.
2	1275154.55	2194886.32	2922.720	2" A.C.
3	1274720.31	2194796.49	2959.514	2" A.C.
4	1274453.52	2195578.05	2906.396	2" A.C.
5	1275042.42	2195785.01	2924.429	3.25 BLM A.C.

LEGEND

- CP-1 SURVEY CONTROL POINT, SURVEYED NOV. 26, 2024
- TP-3 TEST PIT EXCAVATED ON DEC. 5, 2024
- BH-3 BORE HOLE DRILLED ON DEC. 4, 2024
- DH-1 DH-1 DRILL HOLE DRILLED IN 1965 BY NORTHERN TESTING LABORATORIES

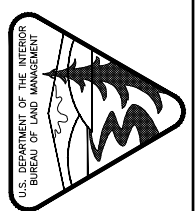
SEE BLM CONTRACT FOR ENTIRE CONSTRUCTION LIMITS. COORDINATE TABLE FOR CONSTRUCTION LIMITS IS AVAILABLE UPON REQUEST.

*ICE/*WATER LEVEL IN THE RESERVOIR (~2901.3) AND STILLING BASIN (~2875.8) AT THE TIME OF THE SURVEY. CONTOURS BELOW WATER LEVEL ARE BASED ON RECORD DRAWING INFORMATION AND WERE ASSUMED FOR DESIGN PURPOSES.



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SITE PLAN AND SURVEY CONTROL

BAR ISLAND DAM MODIFICATIONS

MONTANA

MARK	REVISION	DATE	APPROVED

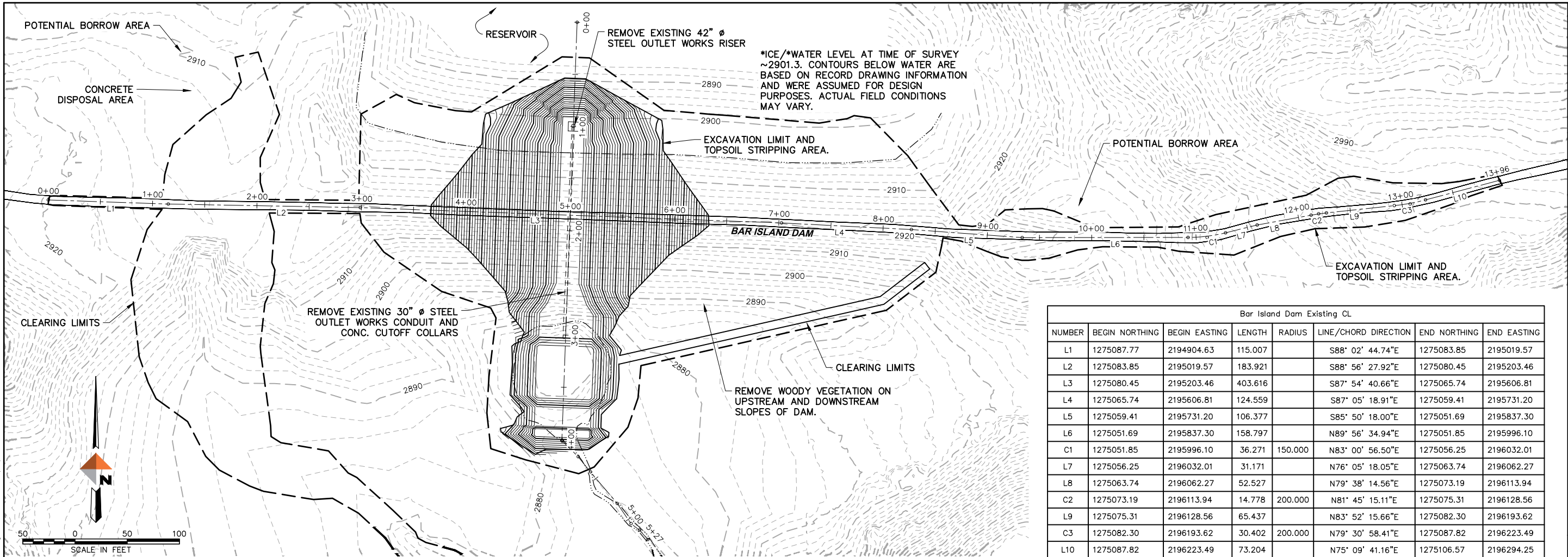
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DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)	DATE: 12-22-25
DESIGNED BY: DAR	DATE: 12-22-25
DRAWN BY: IME	DATE: 12-22-25
CHECKED BY: RCR	DATE: 12-22-25
APPROVED BY: DAR	DATE: 12-22-25

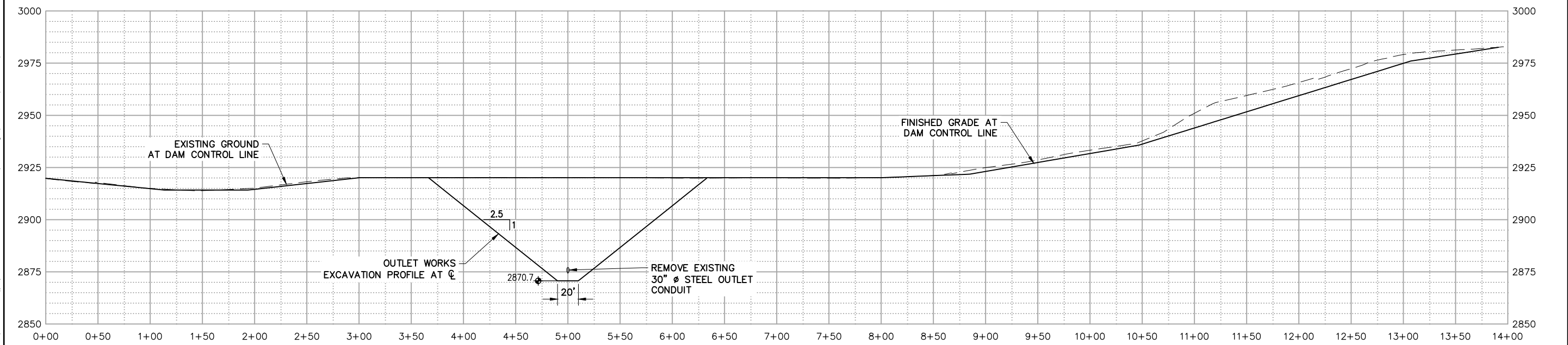
C1

SHEET 5 OF 26

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Bar Island Dam Existing CL							
NUMBER	BEGIN NORTHING	BEGIN EASTING	LENGTH	RADIUS	LINE/CHORD DIRECTION	END NORTHING	END EASTING
L1	1275087.77	2194904.63	115.007		S88° 02' 44.74"E	1275083.85	2195019.57
L2	1275083.85	2195019.57	183.921		S88° 56' 27.92"E	1275080.45	2195203.46
L3	1275080.45	2195203.46	403.616		S87° 54' 40.66"E	1275065.74	2195606.81
L4	1275065.74	2195606.81	124.559		S87° 05' 18.91"E	1275059.41	2195731.20
L5	1275059.41	2195731.20	106.377		S85° 50' 18.00"E	1275051.69	2195837.30
L6	1275051.69	2195837.30	158.797		N89° 56' 34.94"E	1275051.85	2195996.10
C1	1275051.85	2195996.10	36.271	150.000	N83° 00' 56.50"E	1275056.25	2196032.01
L7	1275056.25	2196032.01	31.171		N76° 05' 18.05"E	1275063.74	2196062.27
L8	1275063.74	2196062.27	52.527		N79° 38' 14.56"E	1275073.19	2196113.94
C2	1275073.19	2196113.94	14.778	200.000	N81° 45' 15.11"E	1275075.31	2196128.56
L9	1275075.31	2196128.56	65.437		N83° 52' 15.66"E	1275082.30	2196193.62
C3	1275082.30	2196193.62	30.402	200.000	N79° 30' 58.41"E	1275087.82	2196223.49
L10	1275087.82	2196223.49	73.204		N75° 09' 41.16"E	1275106.57	2196294.25





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**DAM EMBANKMENT EXCAVATION
PLAN AND PROFILE**

BAR ISLAND DAM MODIFICATIONS

AUTOCAD NAME: SC-CS-EMB-EXCV-12652-01.DWG
DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)
DESIGNED BY: DAR
DRAWN BY: IME
CHECKED BY: RCR
APPROVED BY: DAR

DATE: 12-22-25
DATE: 12-22-25
DATE: 12-22-25
DATE: 12-22-25

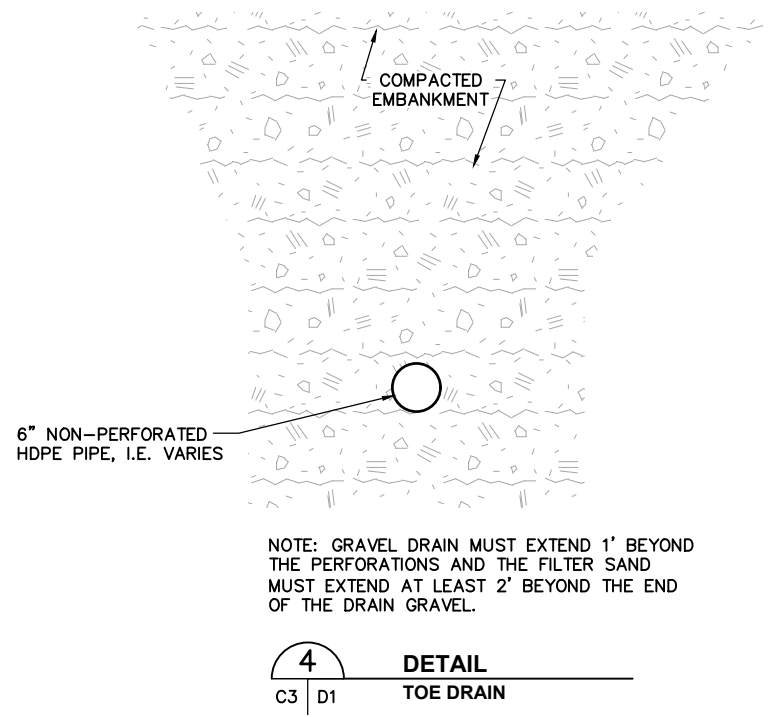
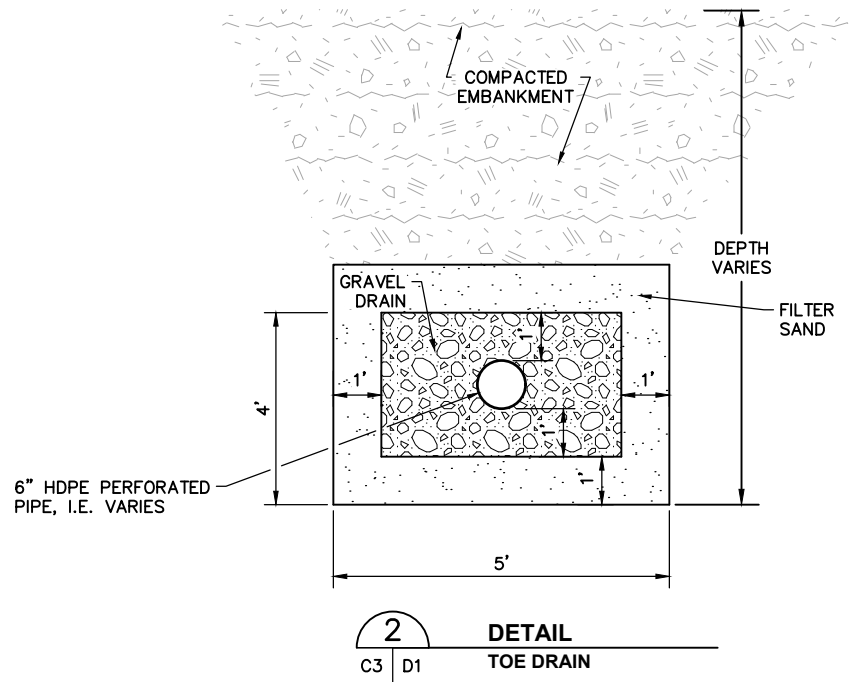
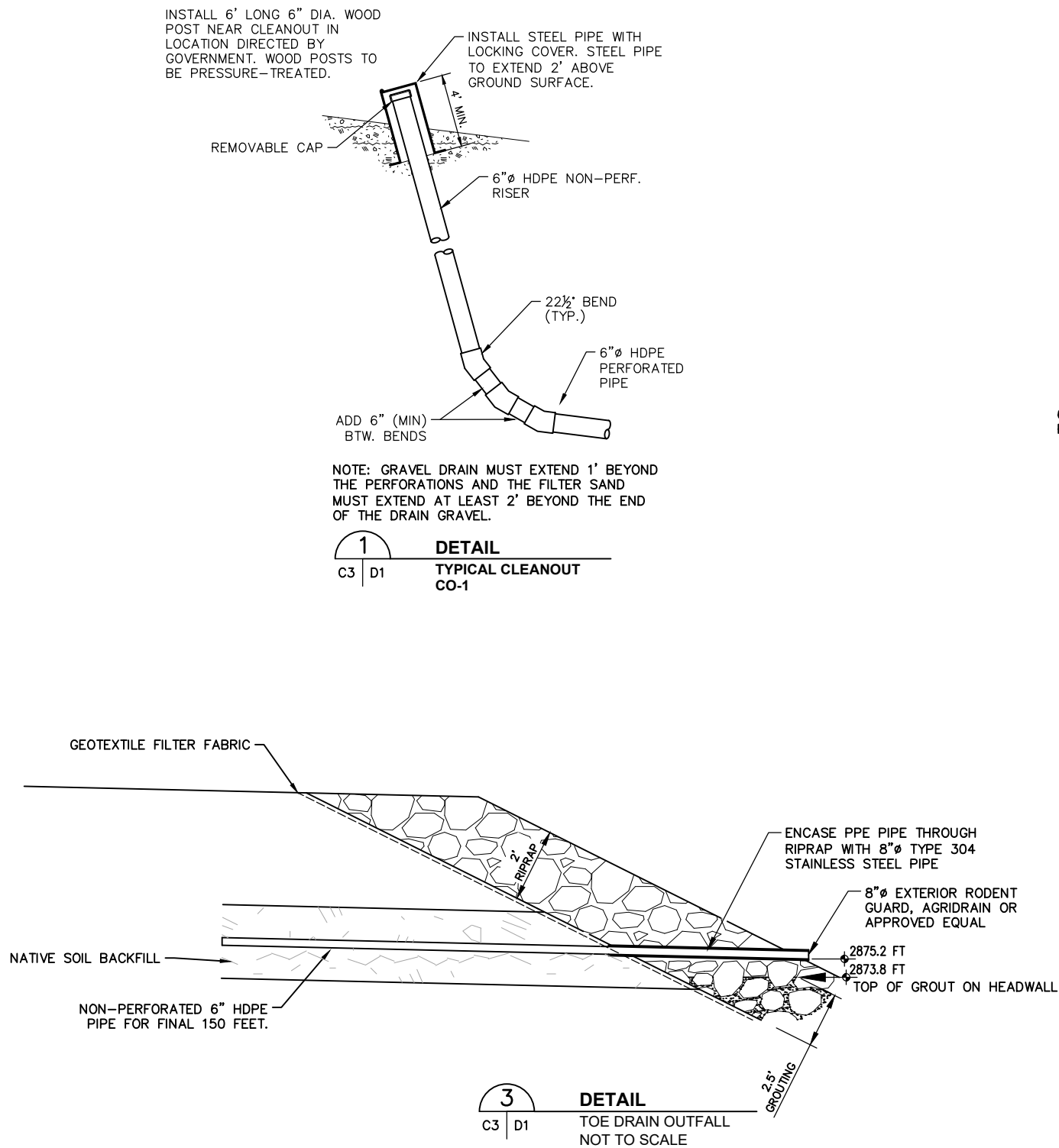
C2

SHEET 6 OF 26

MONTANA

MALTA

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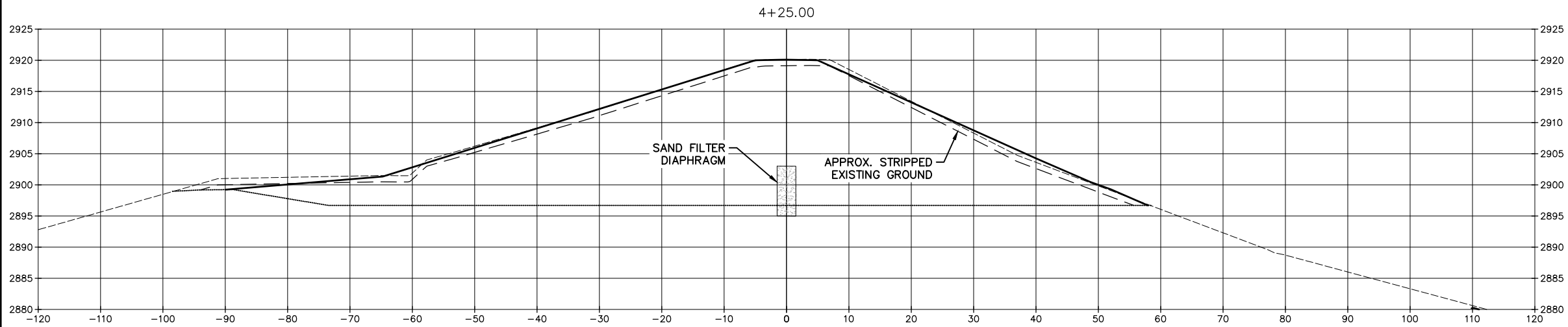
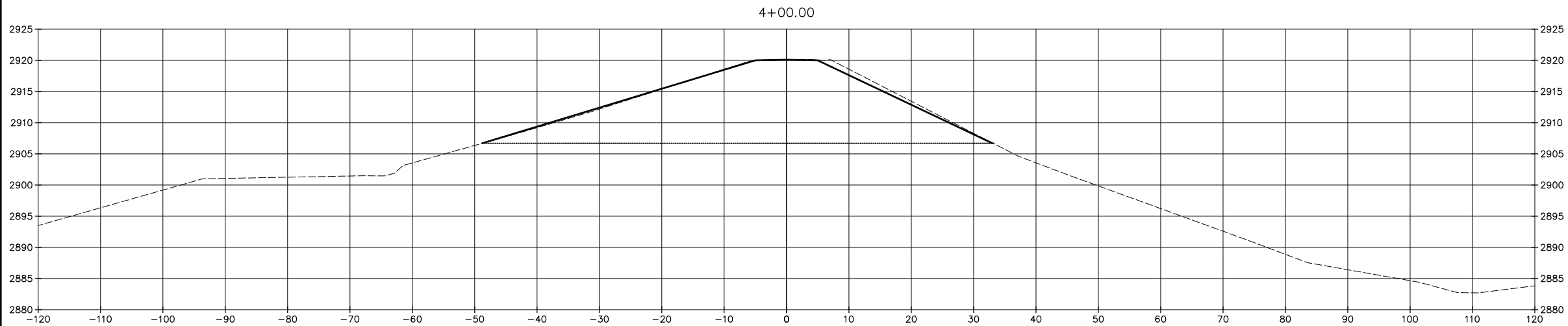
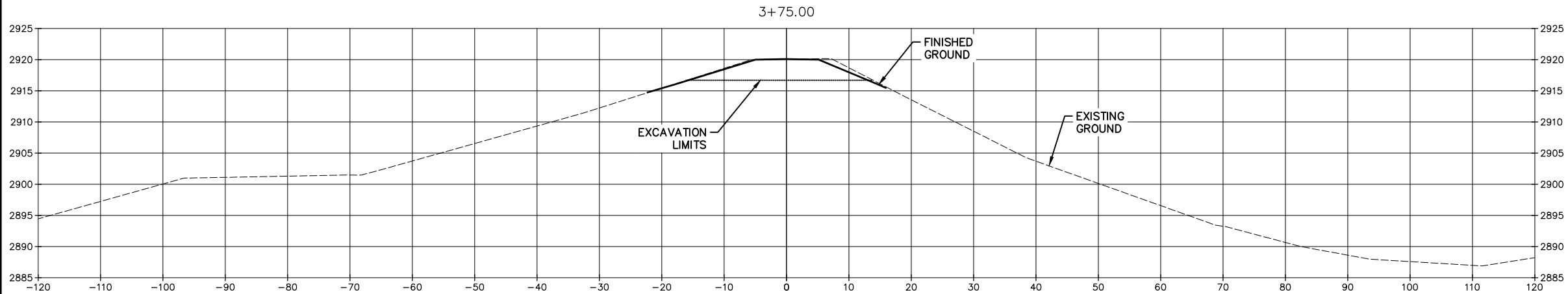
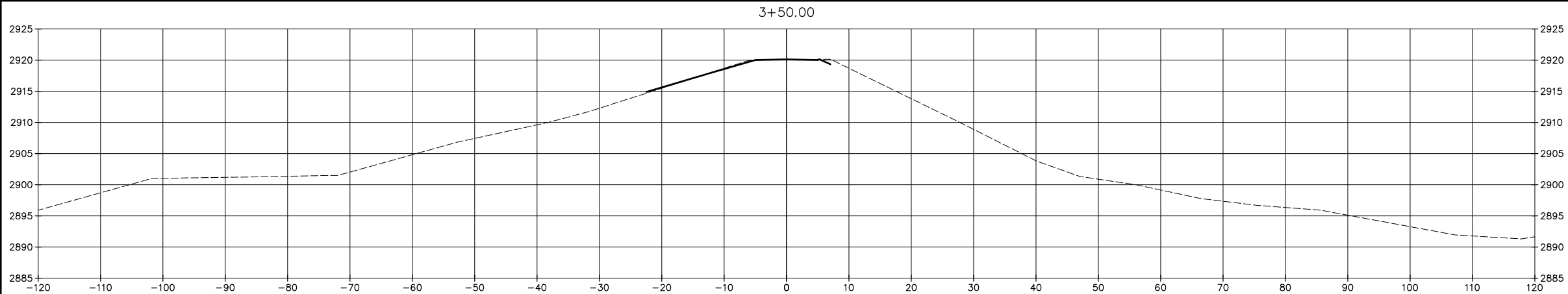
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		UNITED STATES DEPARTMENT OF THE INTERIOR BUREAU OF LAND MANAGEMENT				MONTANA			
		TOE DRAIN DETAILS							
		BAR ISLAND DAM MODIFICATIONS							
		MALTA							
AUTOCAD NAME: SC-CS-EMB-CREST-FG-12652-01.DWG		MARK		REVISION		DATE		APPROVED	
DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)									
DESIGNED BY: DAR		DATE: 12-22-25							
DRAWN BY: IME		DATE: 12-22-25							
CHECKED BY: RCR		DATE: 12-22-25							
APPROVED BY: DAR		DATE: 12-22-25							

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DAM EMBANKMENT SECTIONS
STA. 3+50 TO STA. 4+25

BAR ISLAND DAM MODIFICATIONS

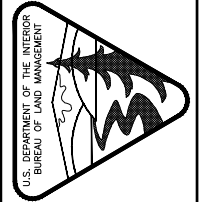
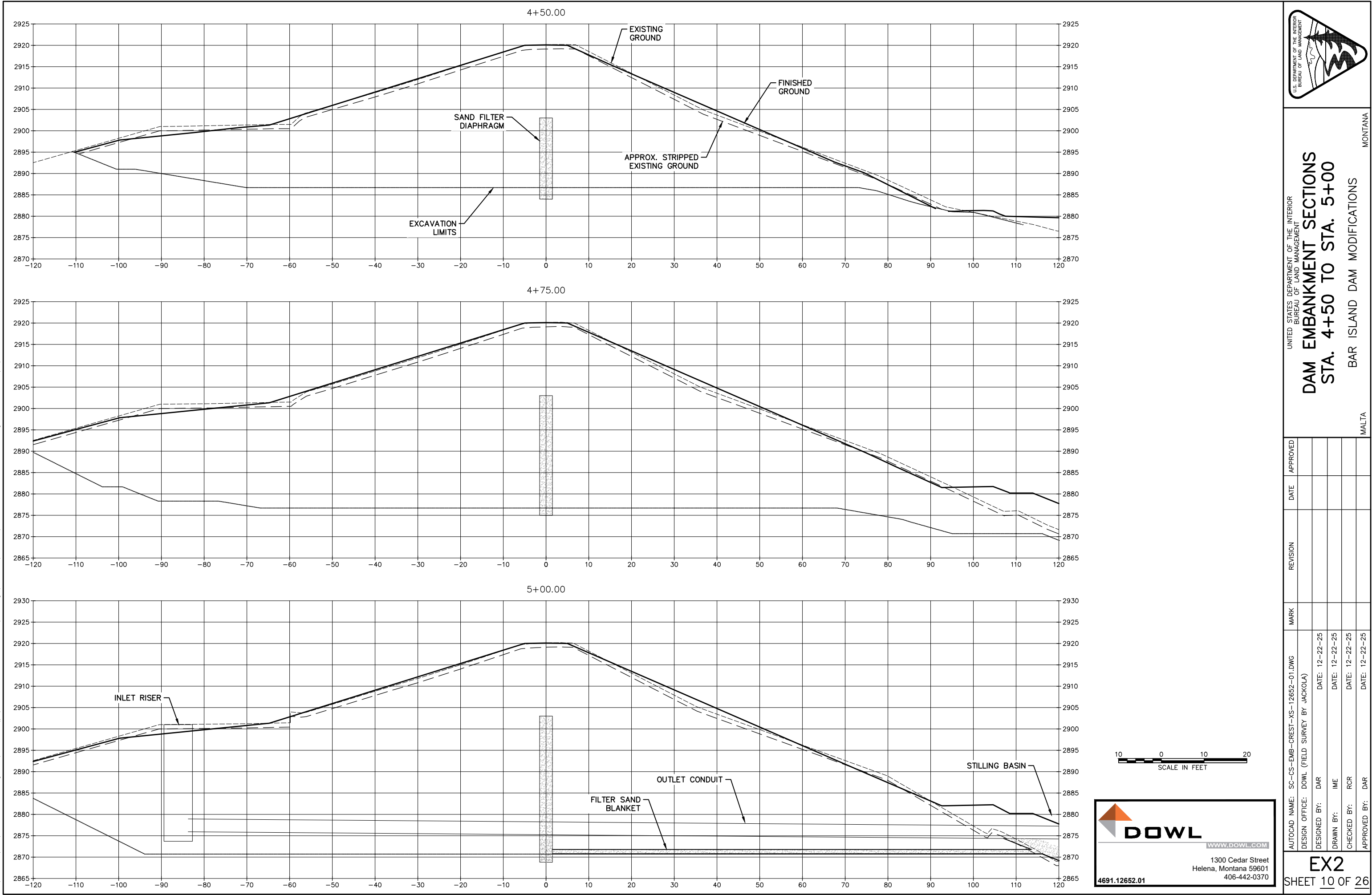
MONTANA

AUTOCAD NAME:	SC-CS-EMB-CREST-XS-12652-01.DWG	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY:	DAR			DATE: 12-22-25	
DRAWN BY:	IME			DATE: 12-22-25	
CHECKED BY:	RCR			DATE: 12-22-25	
APPROVED BY:	DAR			DATE: 12-22-25	

EX1

SHEET 9 OF 26

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DAM EMBANKMENT SECTIONS
STA. 4+50 TO STA. 5+00

BAR ISLAND DAM MODIFICATIONS

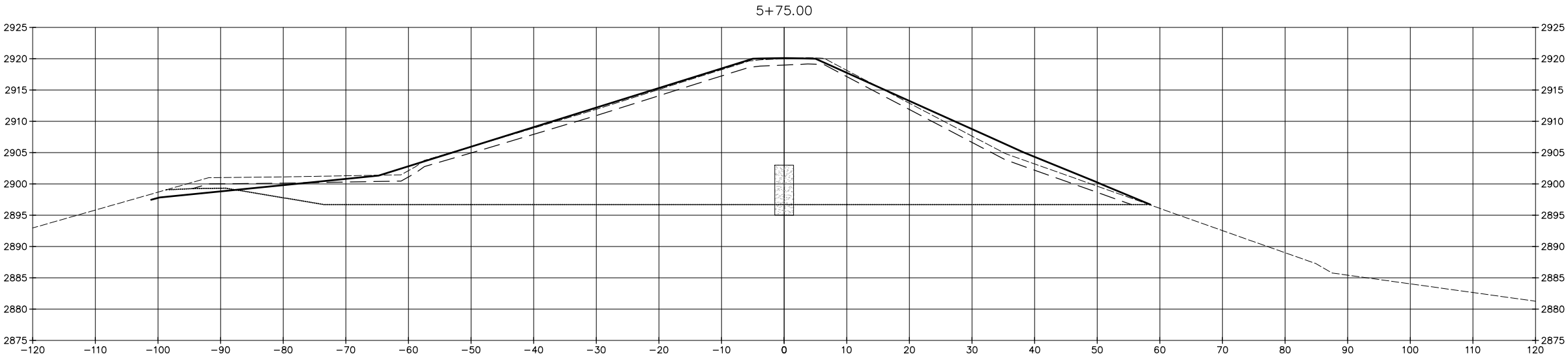
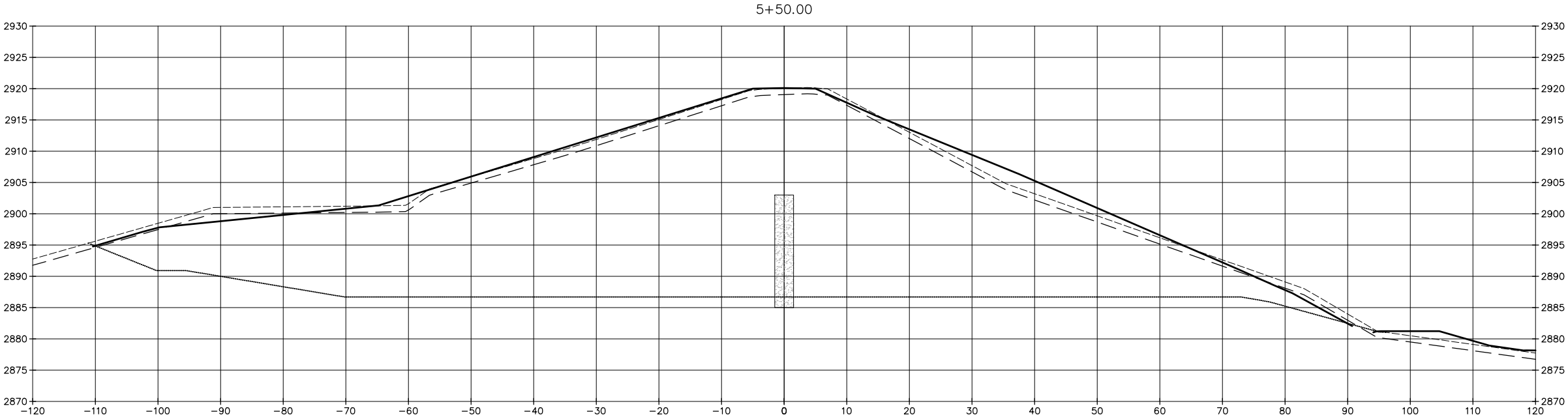
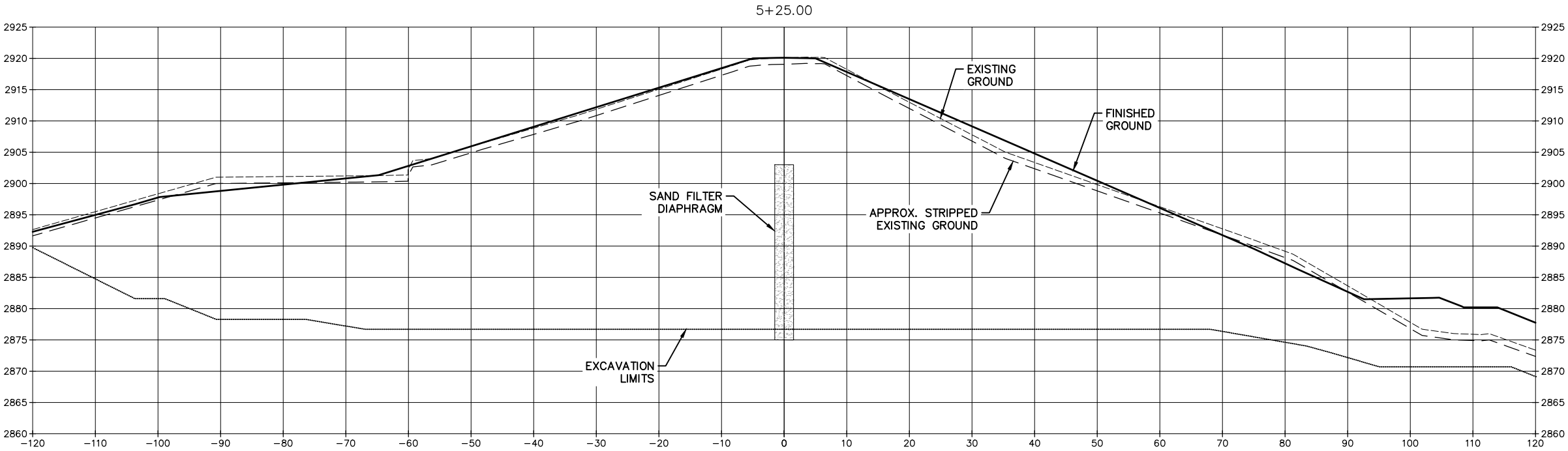
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AUTOCAD NAME:	SC-CS-EMB-CREST-XS-12652-01.DWG	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY:	DAR				
DRAWN BY:	IME				
CHECKED BY:	ROR				
APPROVED BY:	DAR				

EX2

SHEET 10 OF 26

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DAM EMBANKMENT SECTIONS
STA. 5+25 TO STA. 5+75
BAR ISLAND DAM MODIFICATIONS

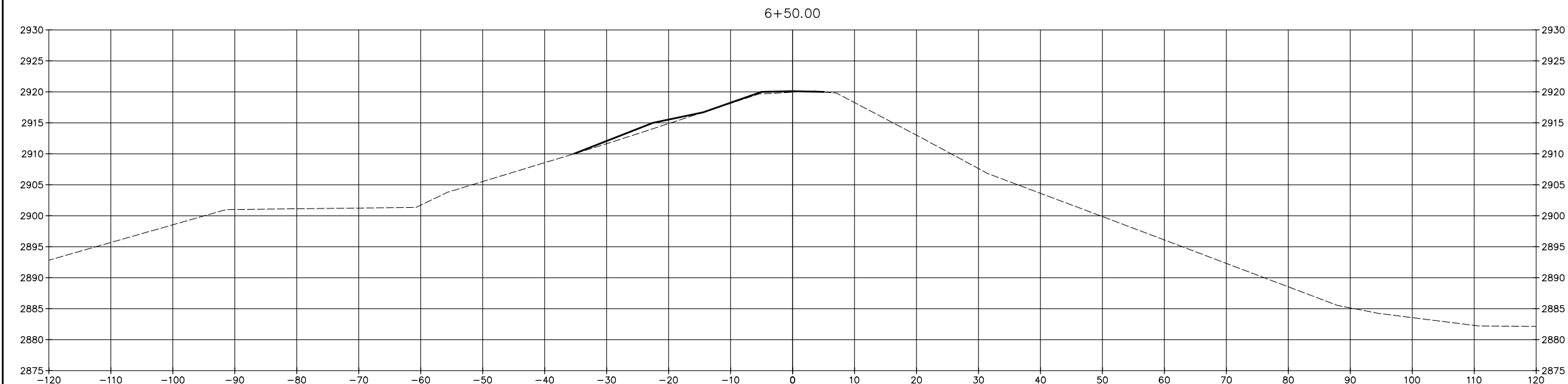
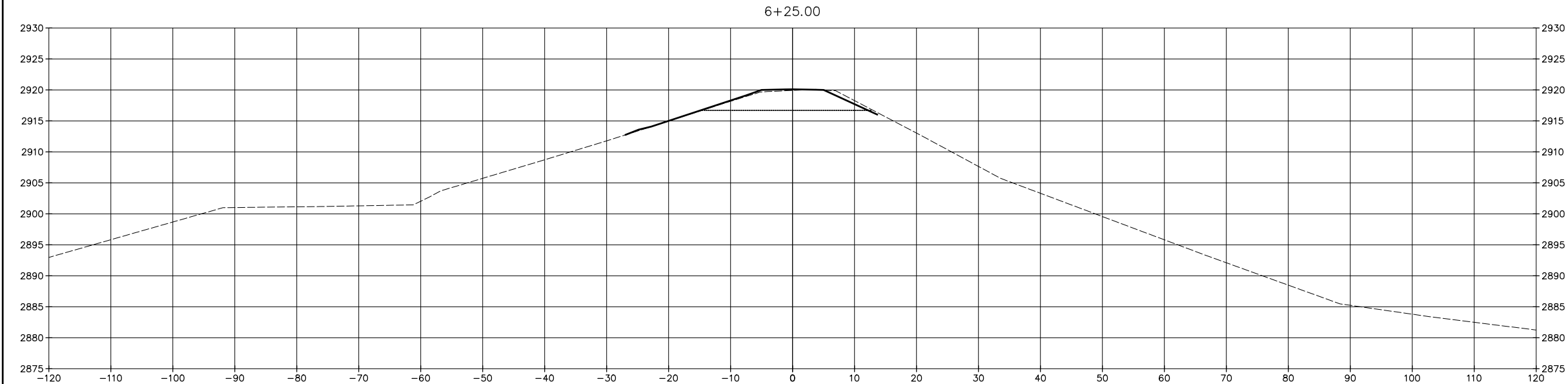
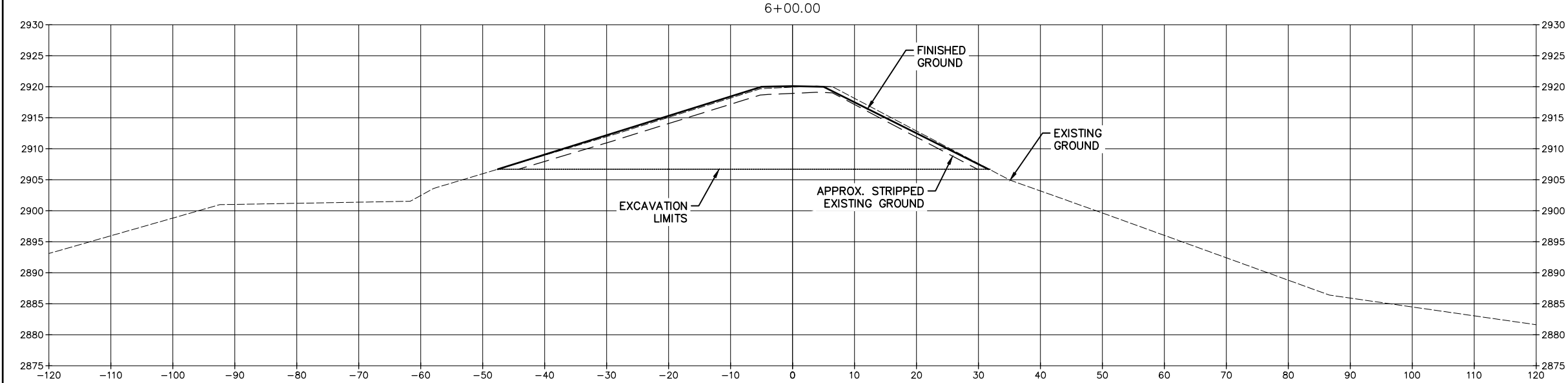
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MARK	REVISION	DATE	APPROVED

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DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)
DESIGNED BY: DAR DATE: 12-22-25
DRAWN BY: IME DATE: 12-22-25
CHECKED BY: RCR DATE: 12-22-25
APPROVED BY: DAR DATE: 12-22-25

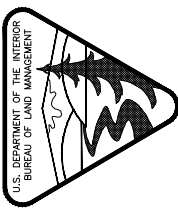
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SHEET 11 OF 26

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10 0 10 20
SCALE IN FEET





UNITED STATES DEPARTMENT OF THE INTERIOR
BUREAU OF LAND MANAGEMENT

DAM EMBANKMENT SECTIONS

STA. 6+00 TO STA. 6+50

BAR ISLAND DAM MODIFICATIONS

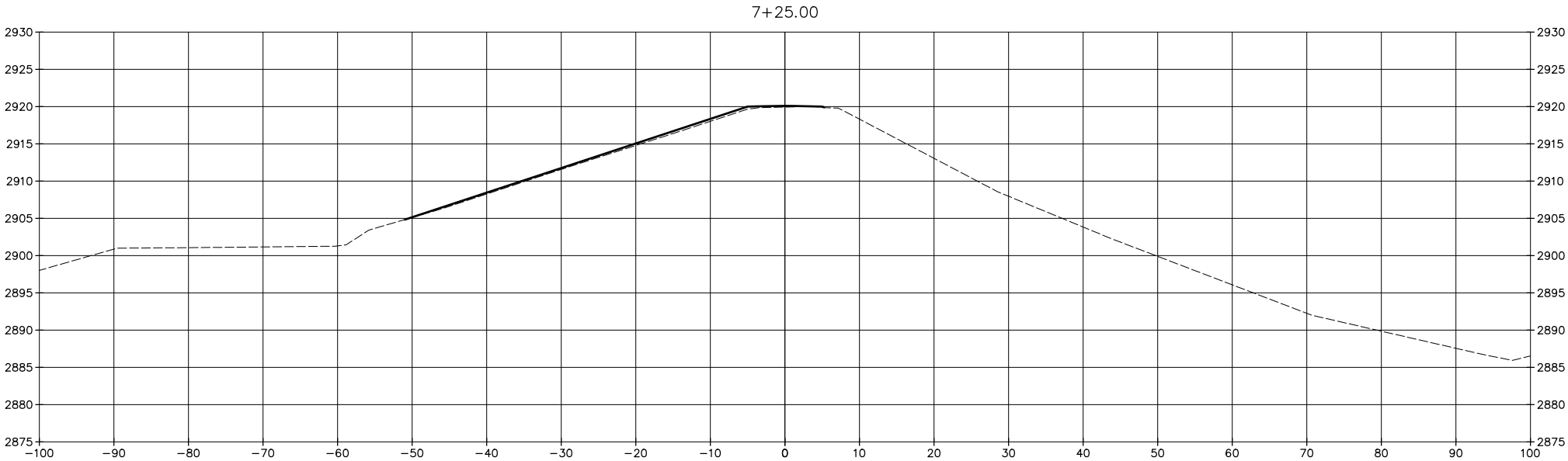
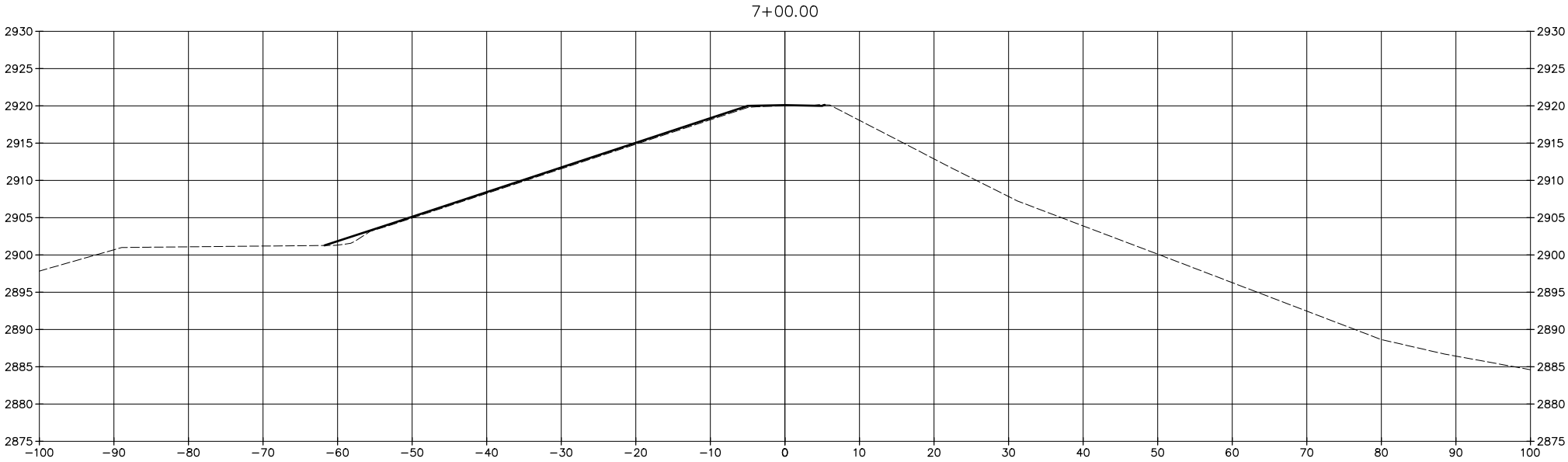
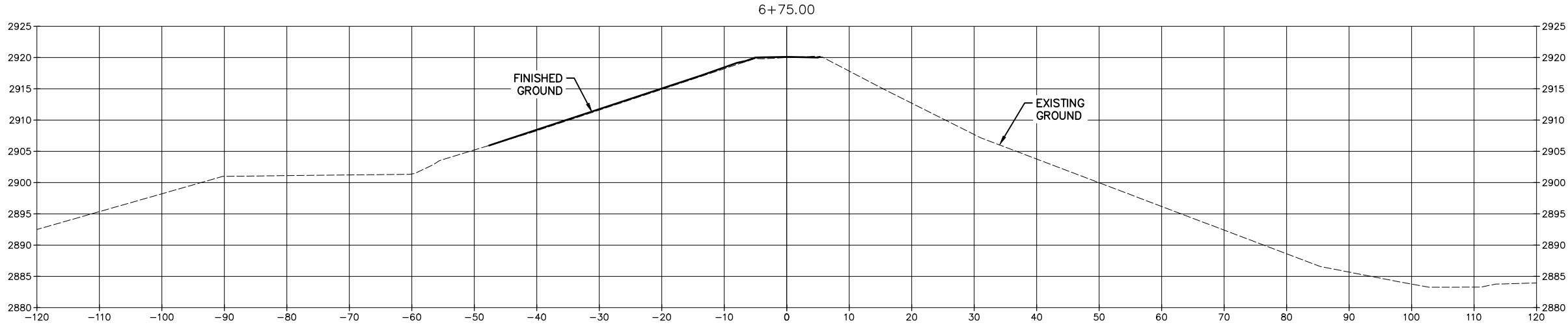
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DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY:	DAR			DATE: 12-22-25	
DRAWN BY:	IME			DATE: 12-22-25	
CHECKED BY:	ROR			DATE: 12-22-25	
APPROVED BY:	DAR			DATE: 12-22-25	

EX4

SHEET 12 OF 26

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UNITED STATES DEPARTMENT OF THE INTERIOR
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DAM EMBANKMENT SECTIONS

STA. 6+75 TO STA. 7+25

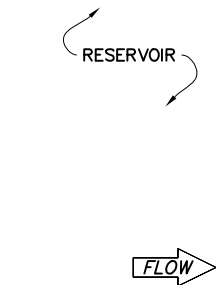
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DESIGNED BY:	DAR				
DRAWN BY:	IME				
CHECKED BY:	ROR				
APPROVED BY:	DAR				

EX5

SHEET 13 OF 26

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0+00

L1

1+00

5+00

2+00

3+00

4+00

REMOVE EXISTING
OUTLET WORKS, PIPE,
AND STRUCTURES

OUTLET WORKS C
STA. 1+85 = DAM C STA. 5+00

EXCAVATION LIMITS

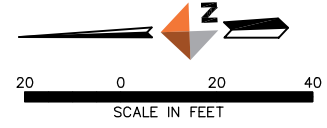
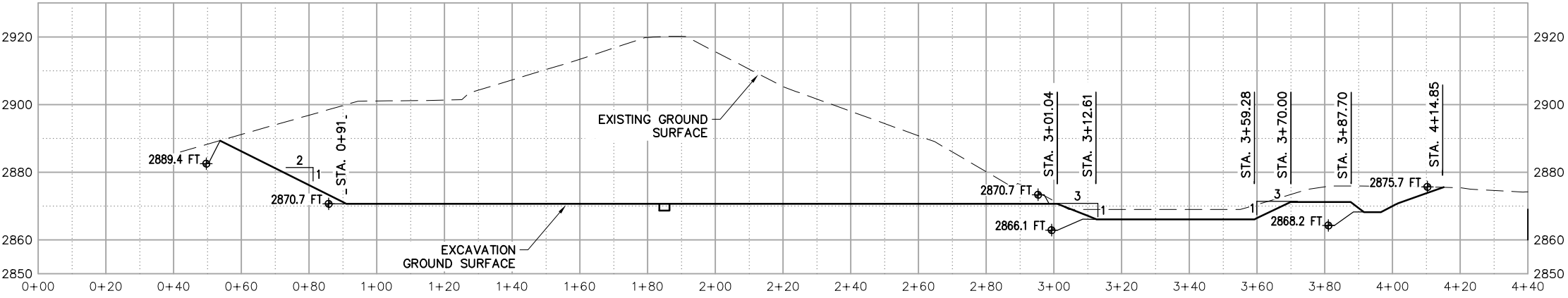
BH-3

BH-2

BH-1

5+00

5+27





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UNITED STATES DEPARTMENT OF THE INTERIOR
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OUTLET WORKS DEMOLITION AND EXCAVATION PLAN

BAR ISLAND DAM MODIFICATIONS

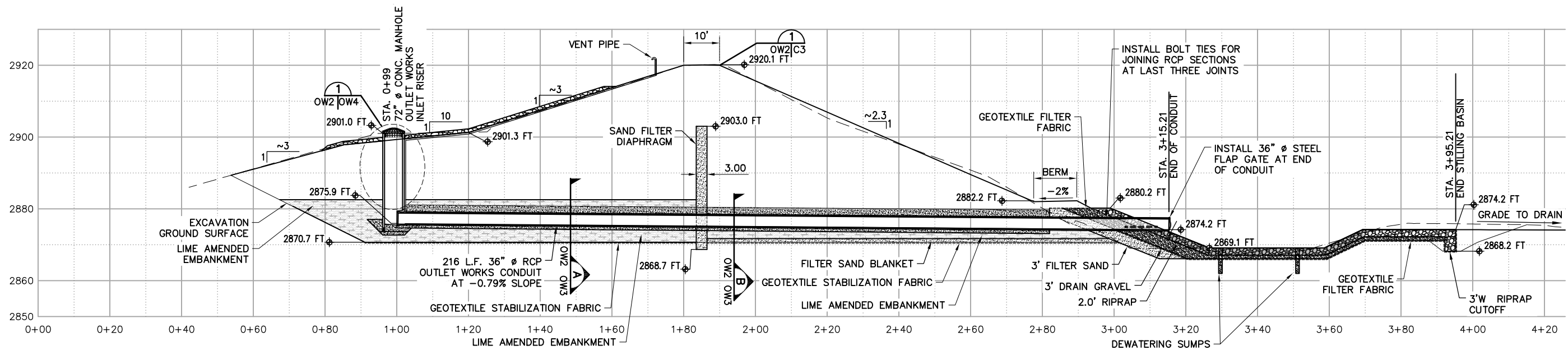
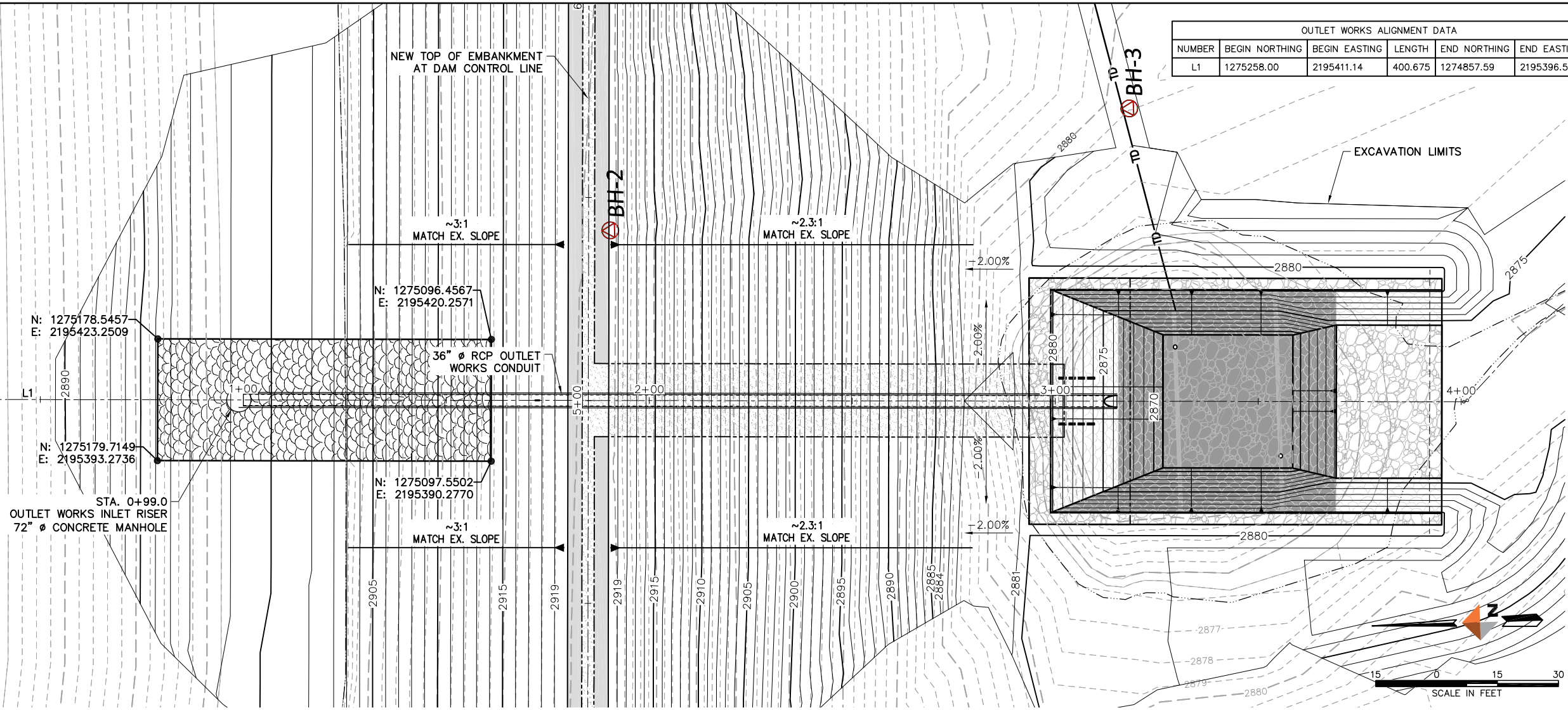
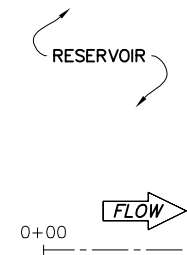
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MALTA

MARK	REVISION	DATE	APPROVED

AUTOCAD NAME: SC-CS-OW-DT-12652-01.DWG	
DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)	
DESIGNED BY: DAR	DATE: 12-22-25
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APPROVED BY: DAR	DATE: 12-22-25

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OUTLET WORKS ALIGNMENT DATA					
NUMBER	BEGIN NORTHING	BEGIN EASTING	LENGTH	END NORTHING	END EASTING
L1	1275258.00	2195411.14	400.675	1274857.59	2195396.54



UNITED STATES DEPARTMENT OF THE INTERIOR
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OUTLET WORKS PLAN AND PROFILE

BAR ISLAND DAM MODIFICATIONS

AUTOCAD NAME:	SC-CS-OW-DT-12652-01.DWG
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)
DESIGNED BY:	DAR
DRAWN BY:	IME
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DATE:	12-22-25
DATE:	12-22-25
DATE:	12-22-25

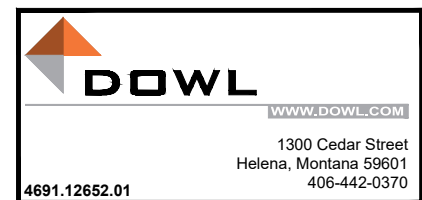
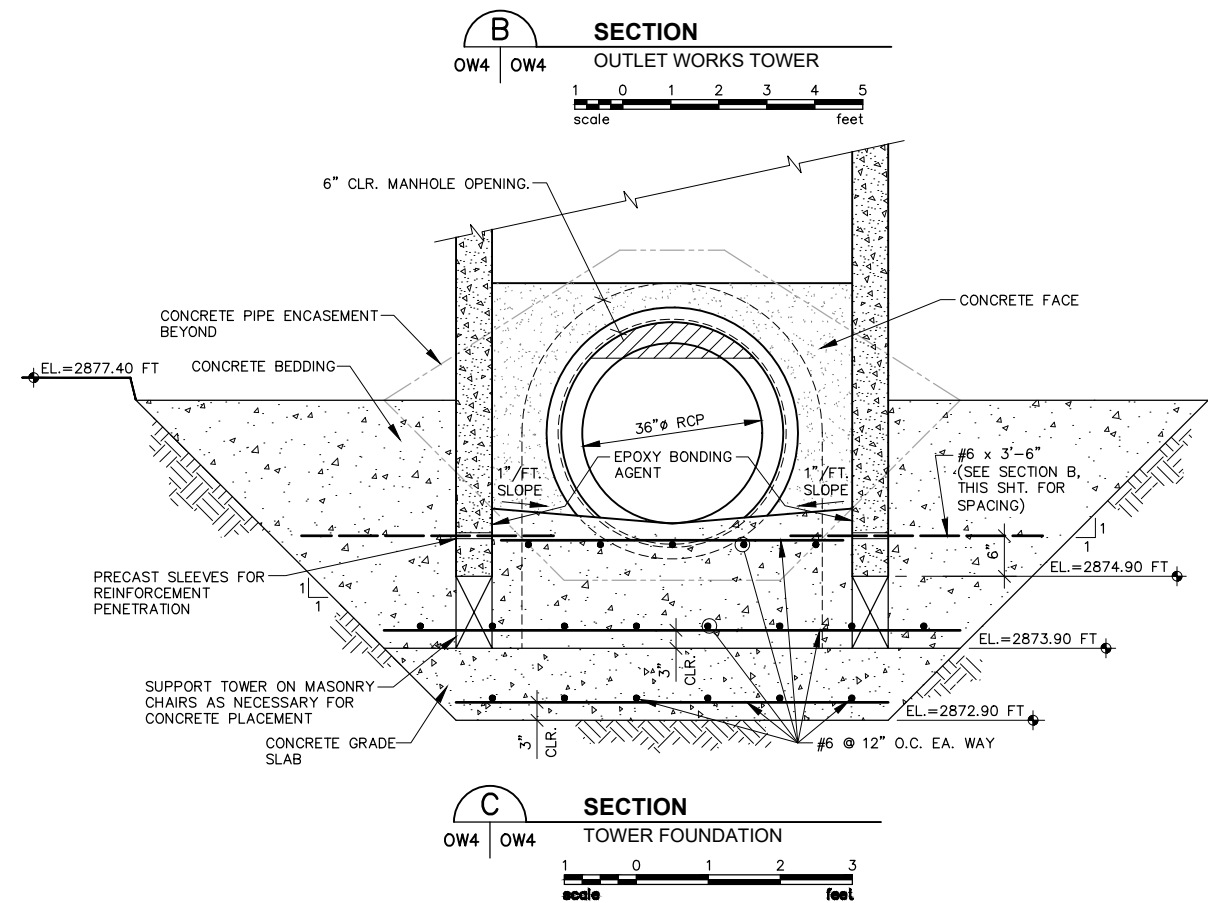
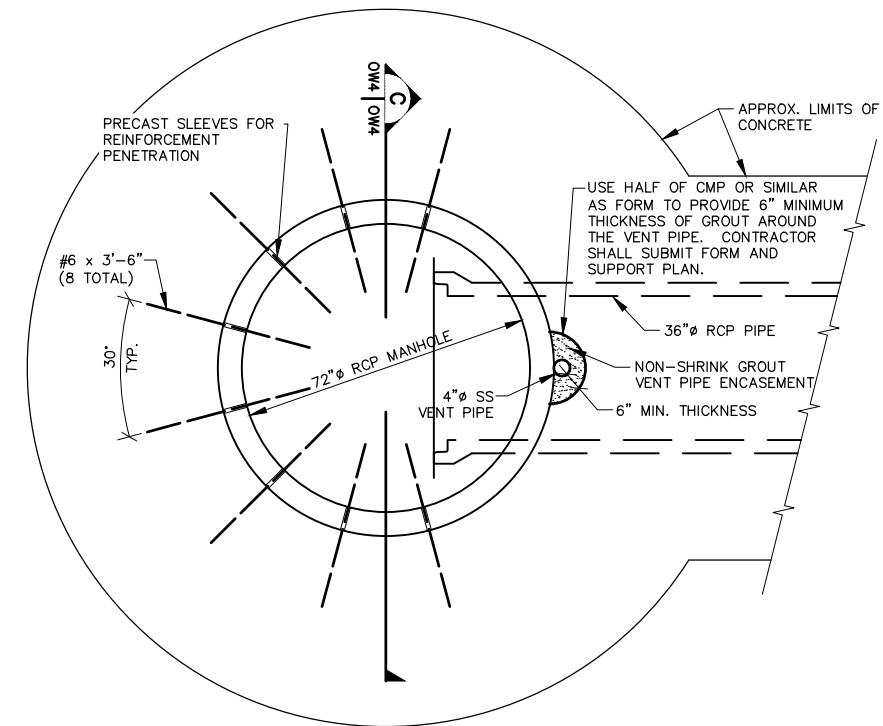
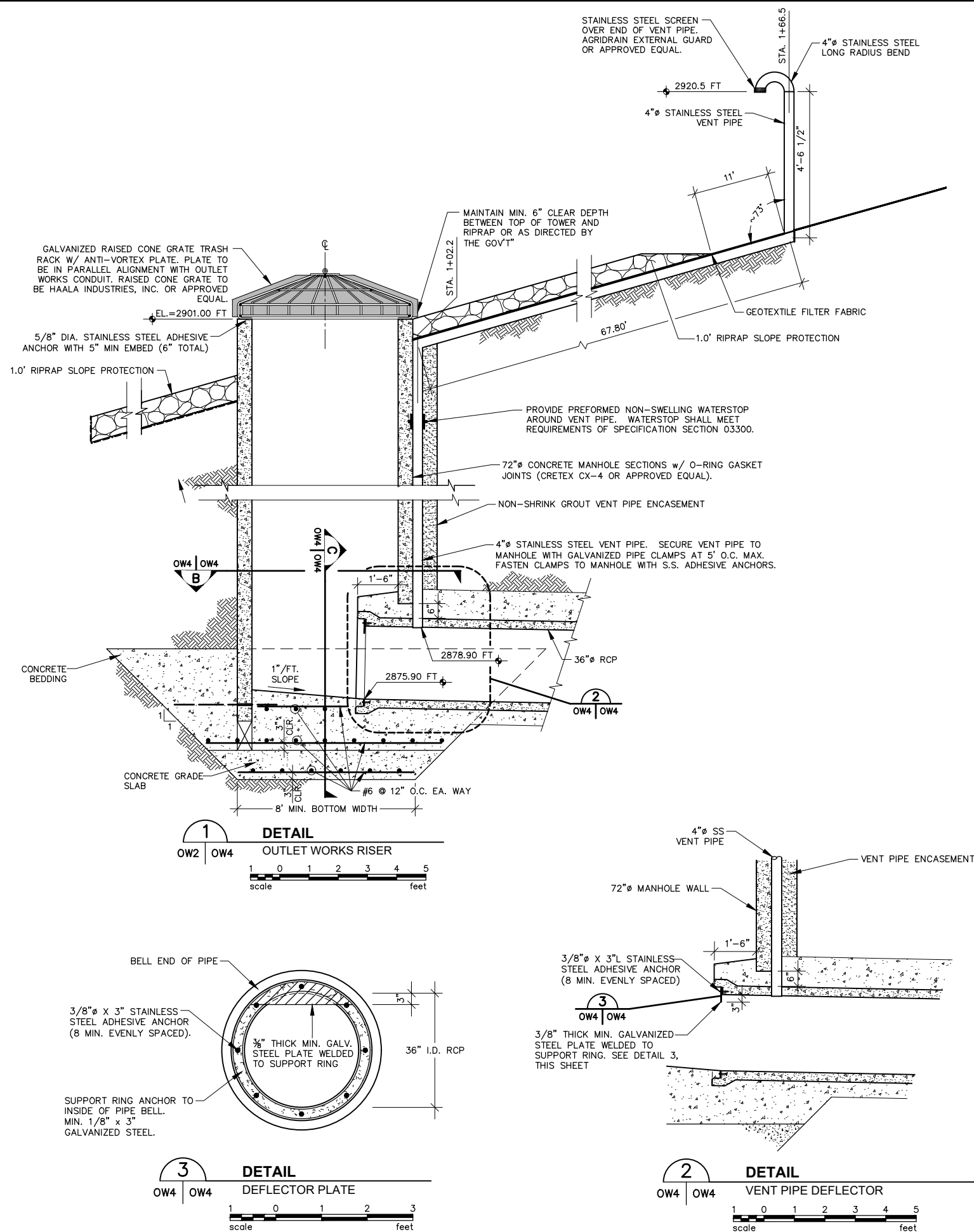


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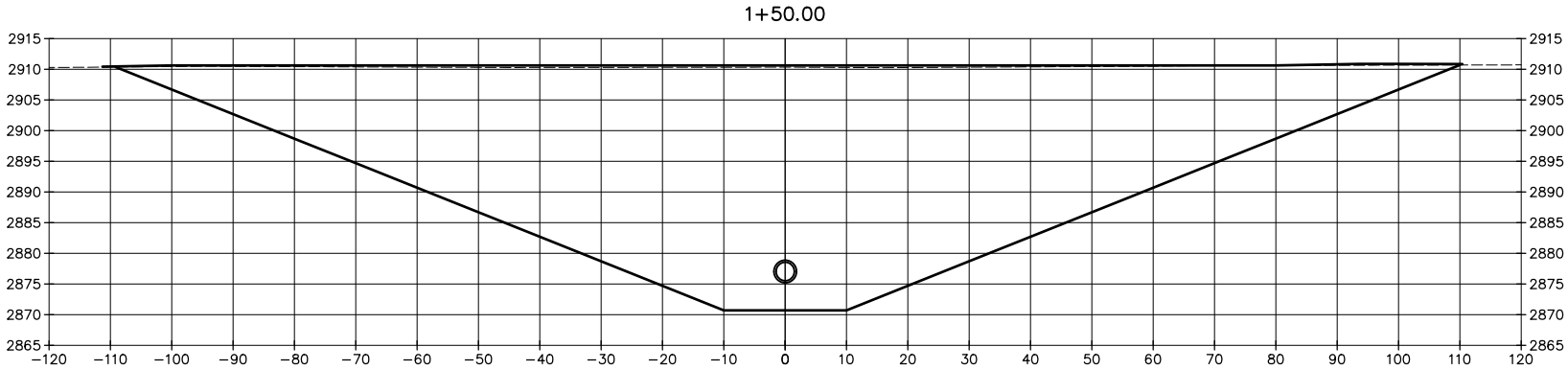
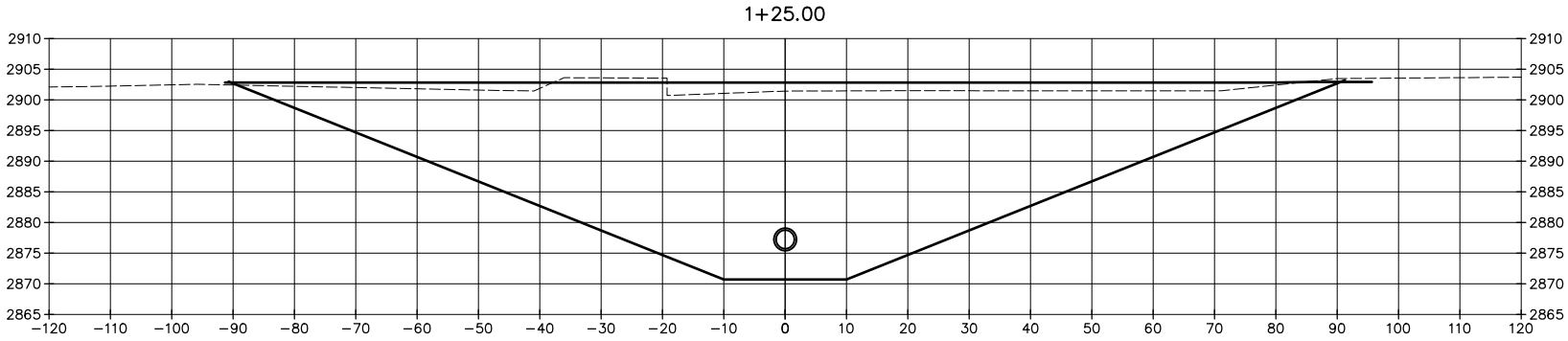
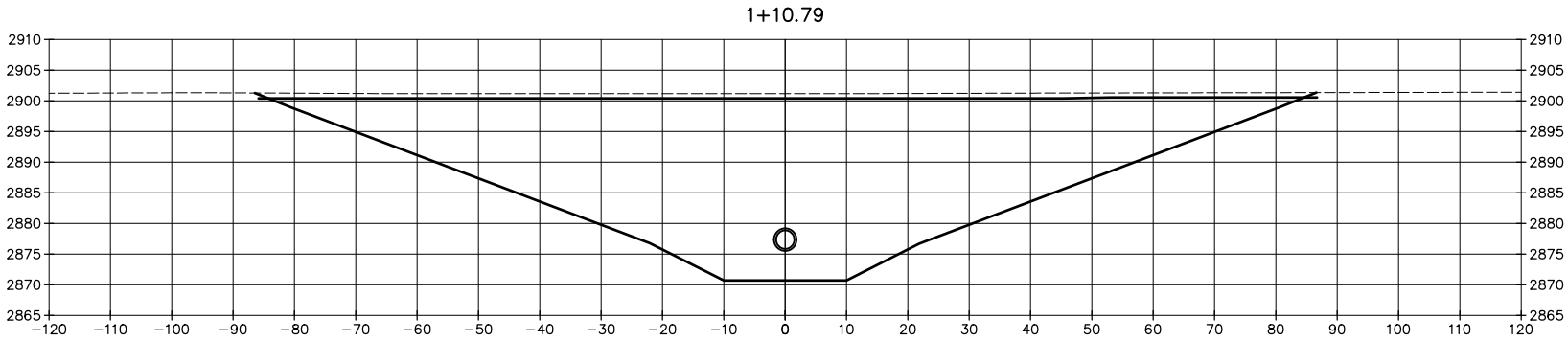
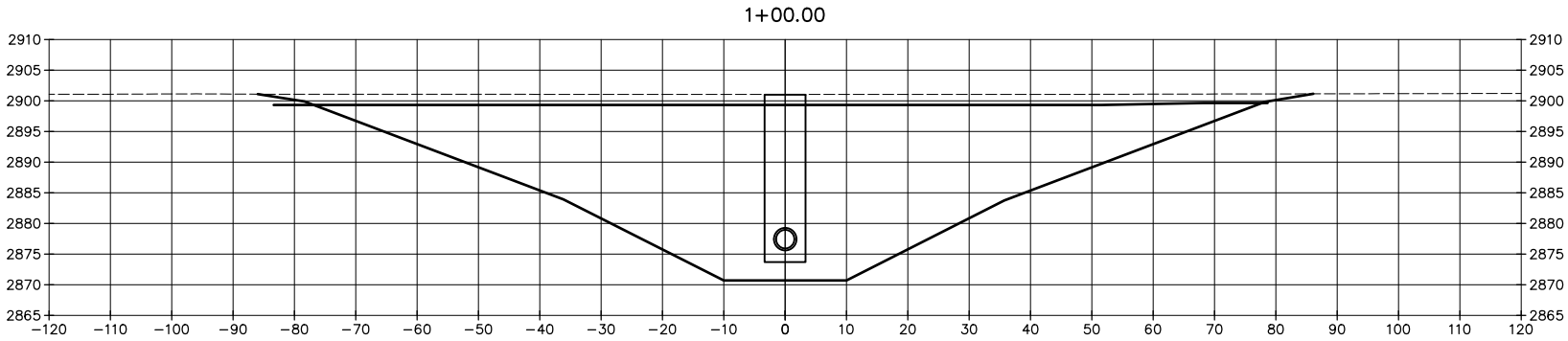
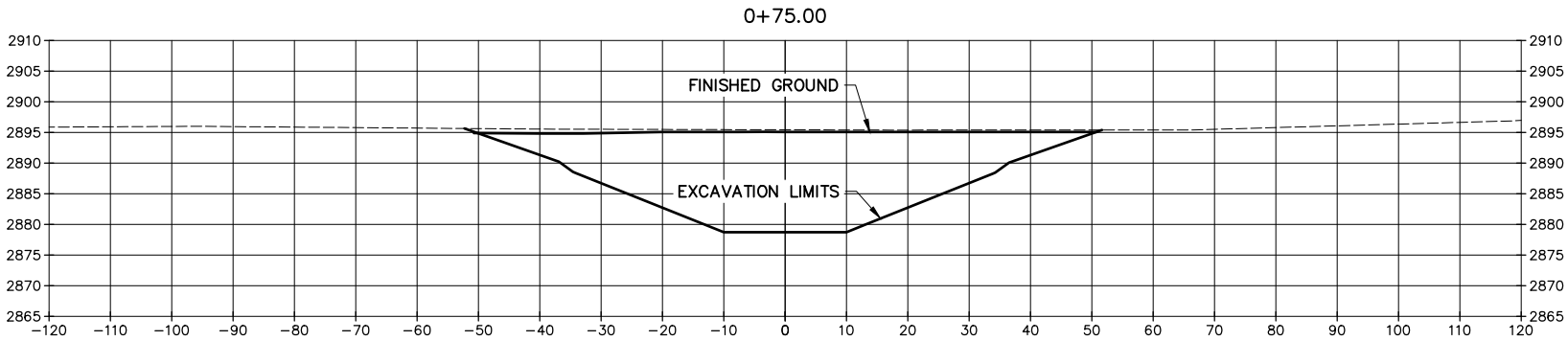
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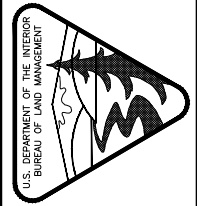


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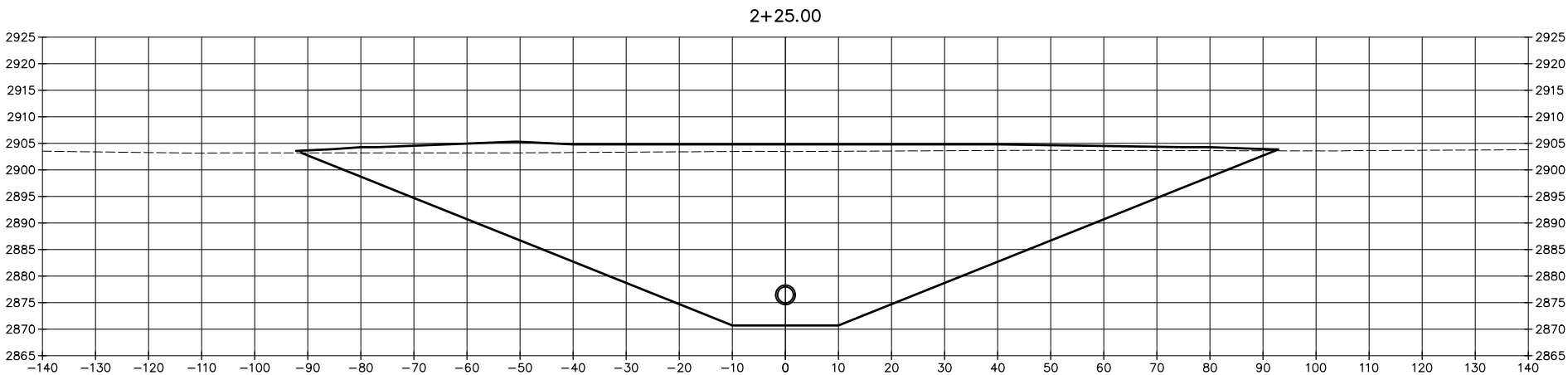
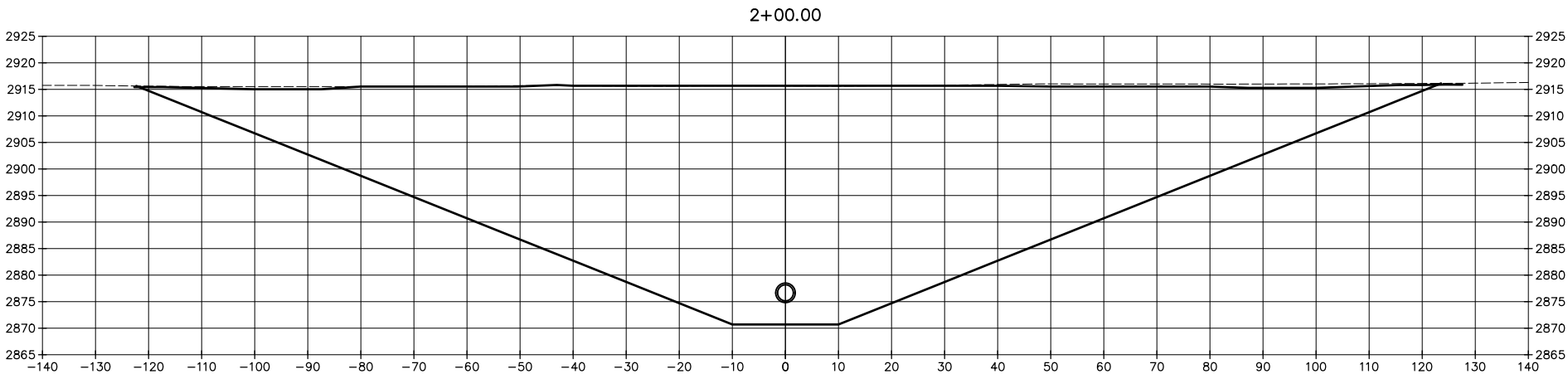
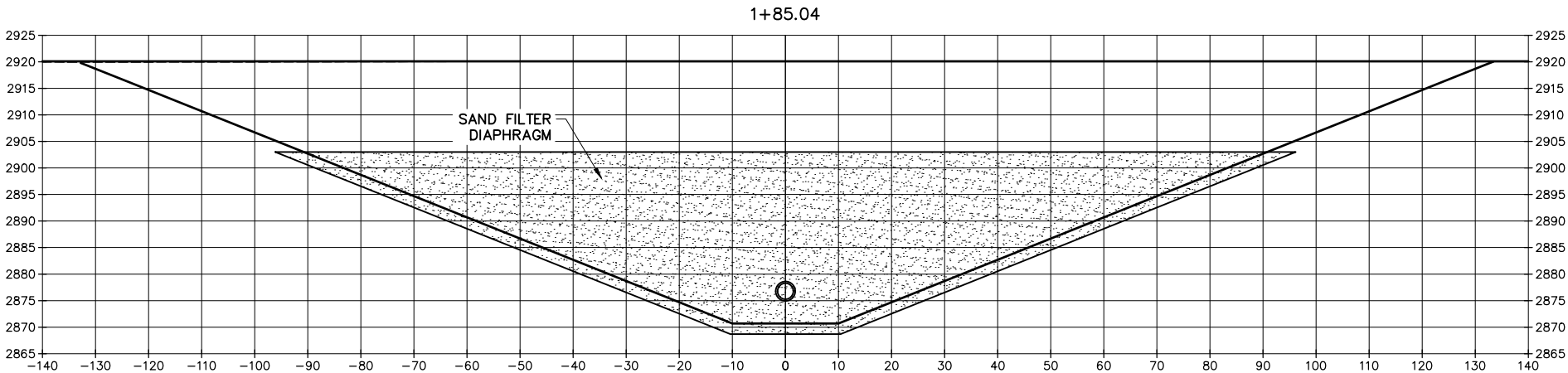
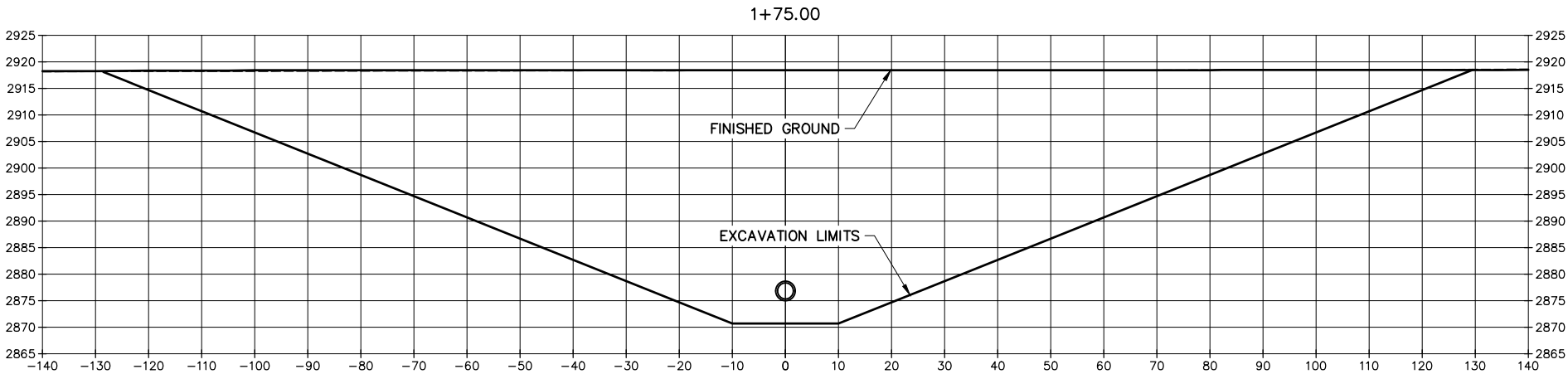
OUTLET WORKS SECTIONS
STA. 0+75 TO STA. 1+50

BAR ISLAND DAM MODIFICATIONS

MONTANA

AUTOCAD NAME:	SC-CS-OW-DT-12652-01.DWG	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY:	DAR			DATE: 12-22-25	
DRAWN BY:	IME			DATE: 12-22-25	
CHECKED BY:	RCR			DATE: 12-22-25	
APPROVED BY:	DAR			DATE: 12-22-25	

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OUTLET WORKS SECTIONS
STA. 1+75 TO STA. 2+25

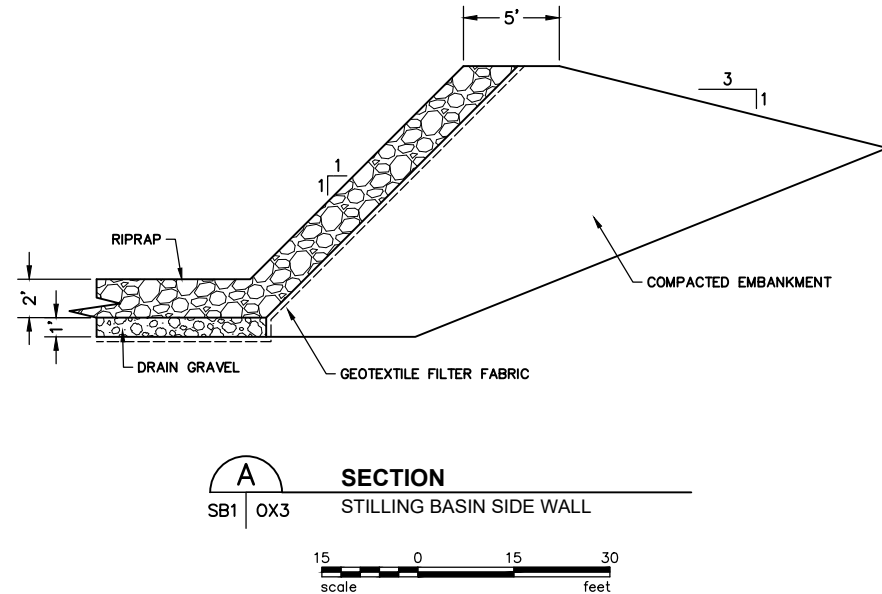
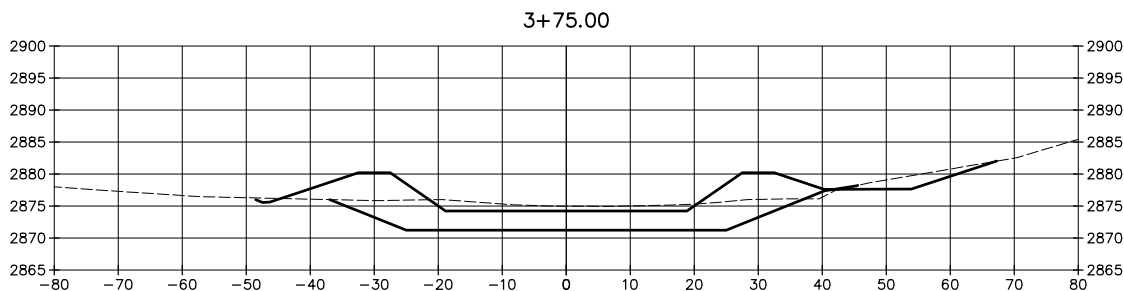
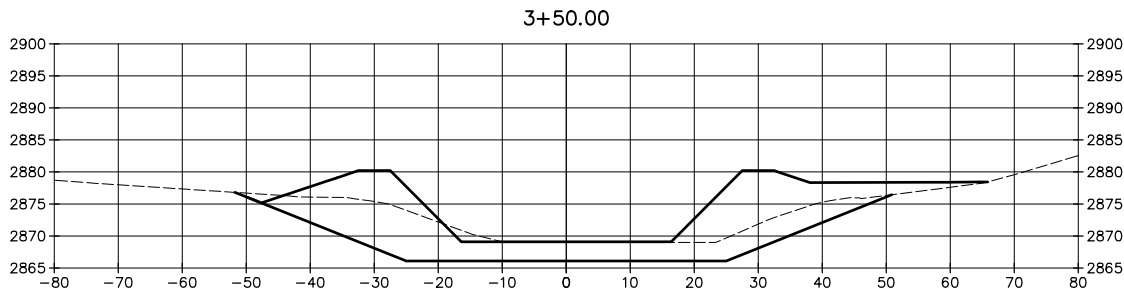
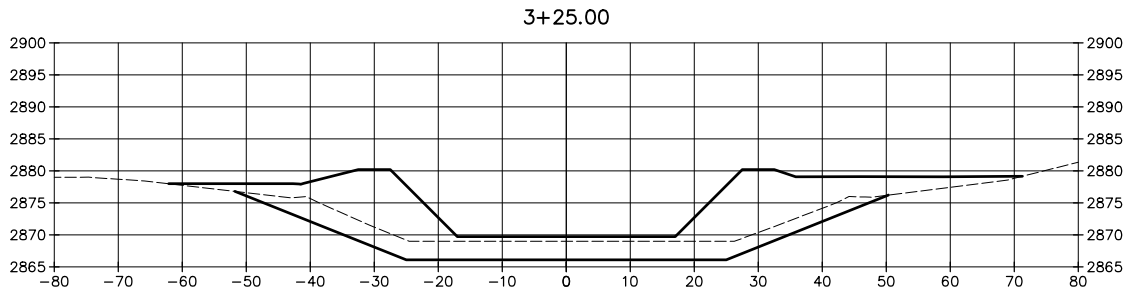
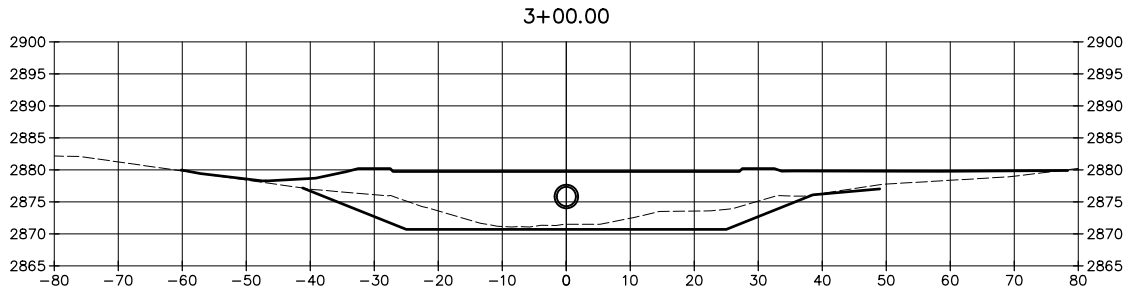
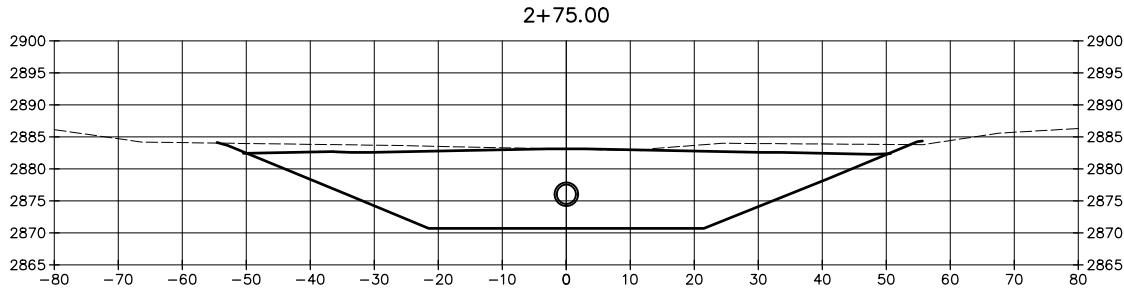
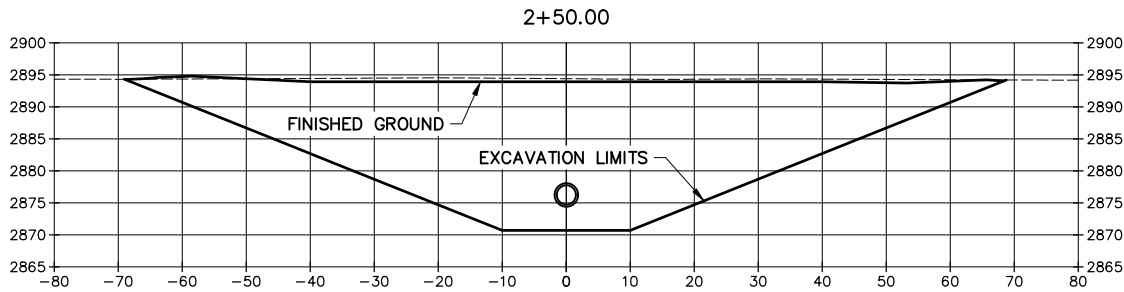
BAR ISLAND DAM MODIFICATIONS

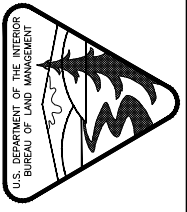
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DESIGNED BY:	DAR			DATE: 12-22-25	
DRAWN BY:	IME			DATE: 12-22-25	
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APPROVED BY:	DAR			DATE: 12-22-25	

OX2
SHEET 19 OF 26

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OUTLET WORKS SECTIONS
STA. 2+50 TO STA. 3+75

BAR ISLAND DAM MODIFICATIONS

MONTANA

MALTA

MARK	REVISION	DATE	APPROVED

AUTOCAD NAME: SC-CS-OW-DT-12652-01.DWG

DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)

DESIGNED BY: DAR DATE: 12-22-25

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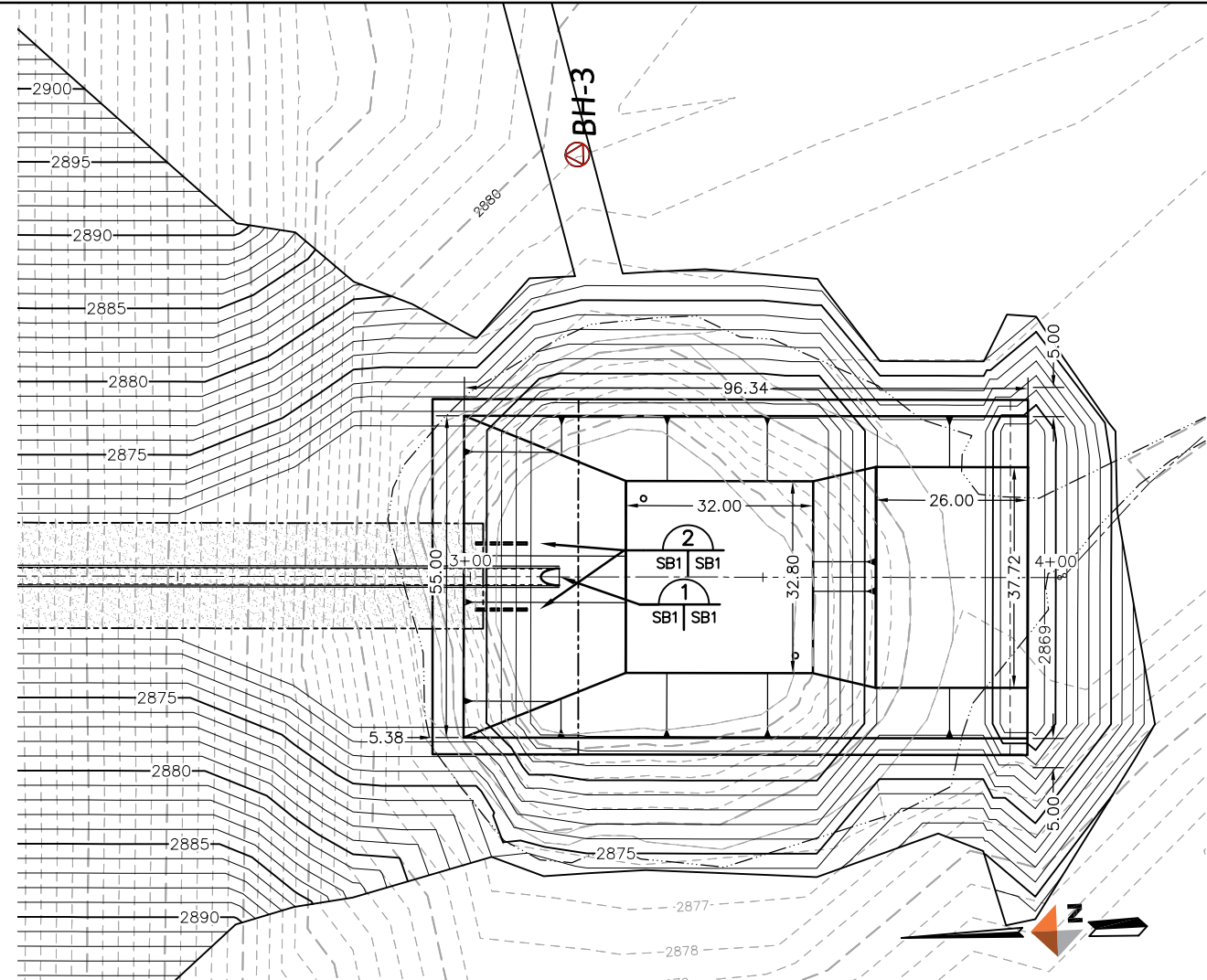
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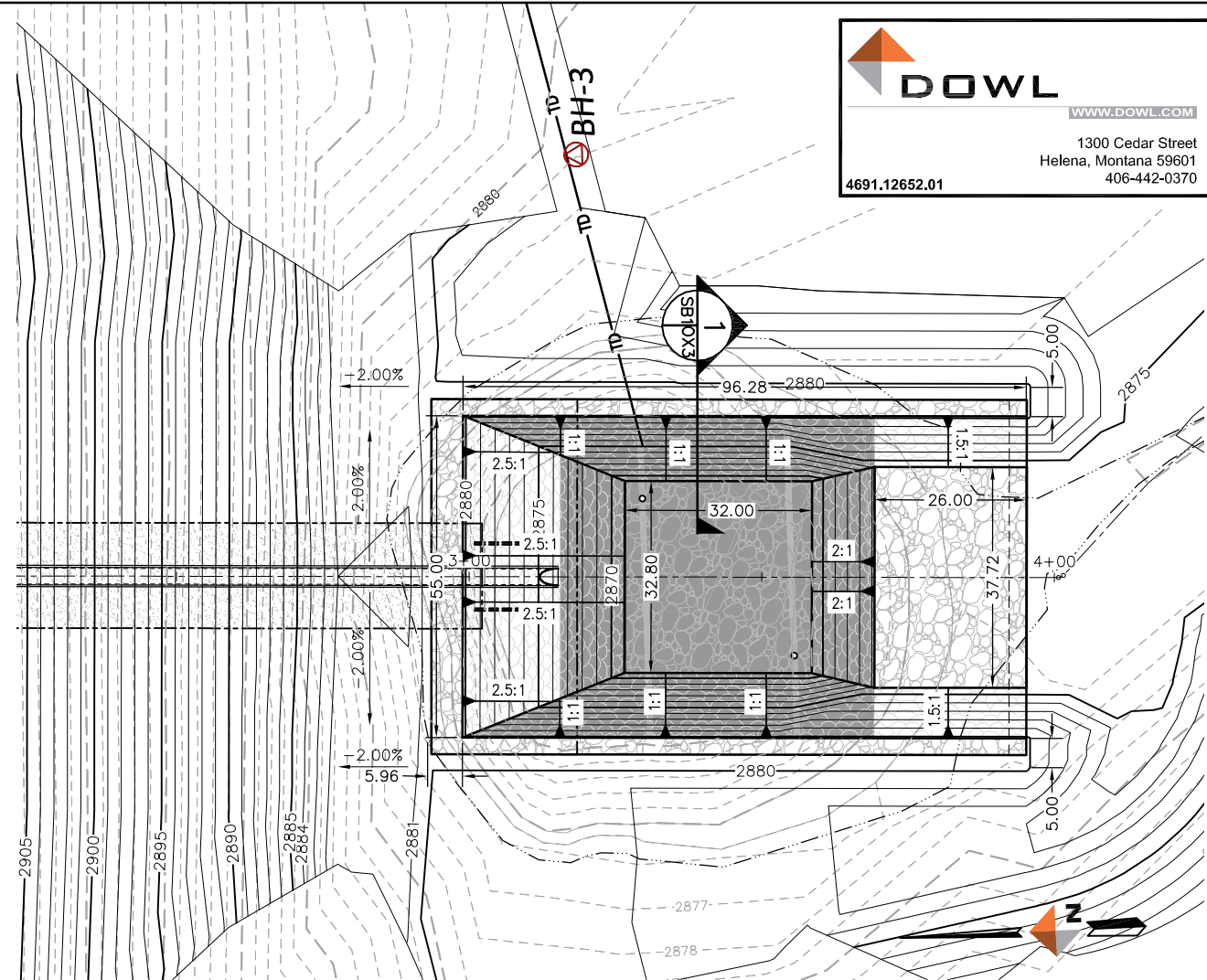
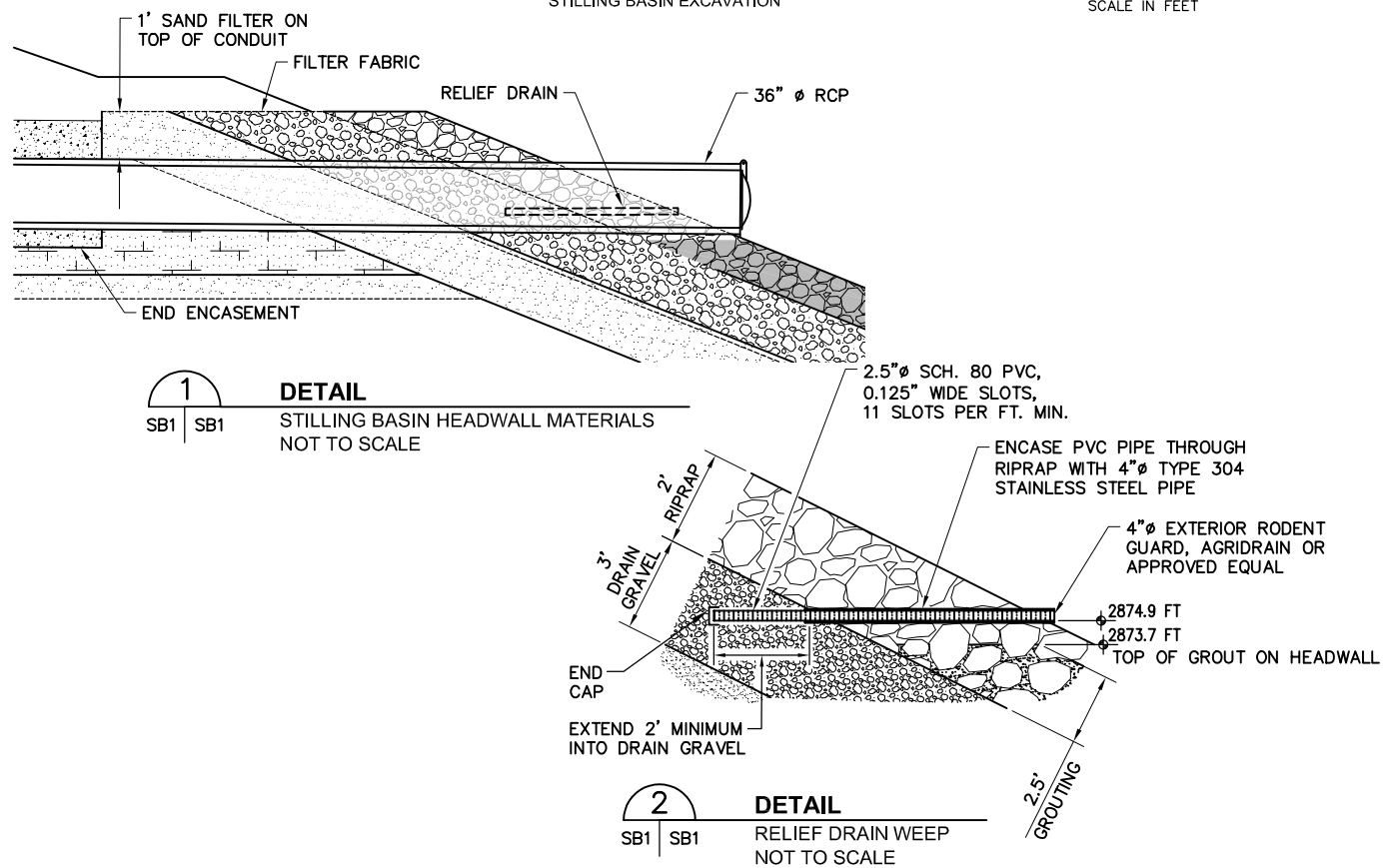
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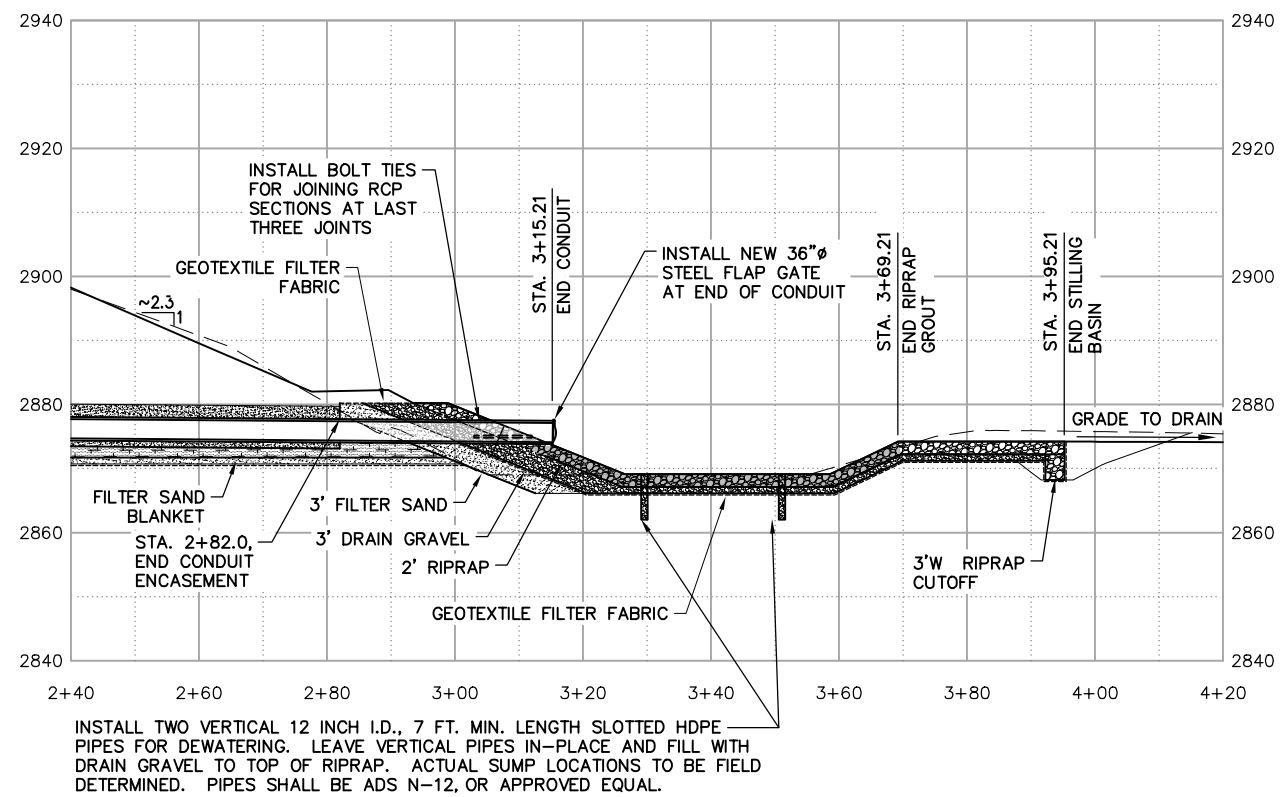
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PLAN
STILLING BASIN EXCAVATION



PLAN
RIPRAP STILLING BASIN





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UNITED STATES DEPARTMENT OF THE INTERIOR
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STILLING BASIN
PLAN AND PROFILE

BAR ISLAND DAM MODIFICATIONS

MONTANA

MARK	REVISION	DATE	APPROVED

AUTOCAD NAME: SC-CS-OW-DT-12652-01.DWG

DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)

DESIGNED BY: DAR

DRAWN BY: IME

CHECKED BY: RCR

APPROVED BY: DAR

DATE: 12-22-25

DATE: 12-22-25

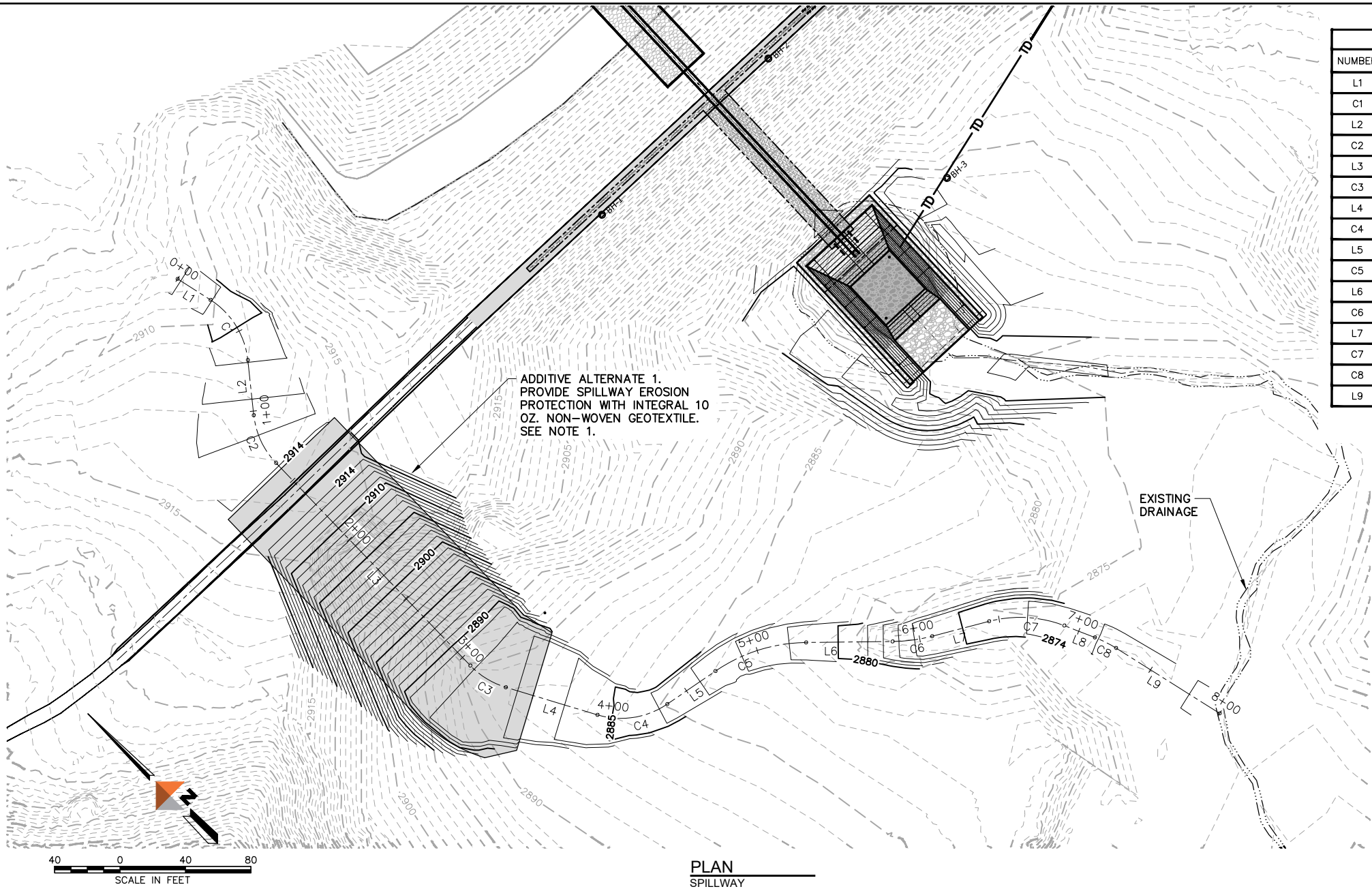
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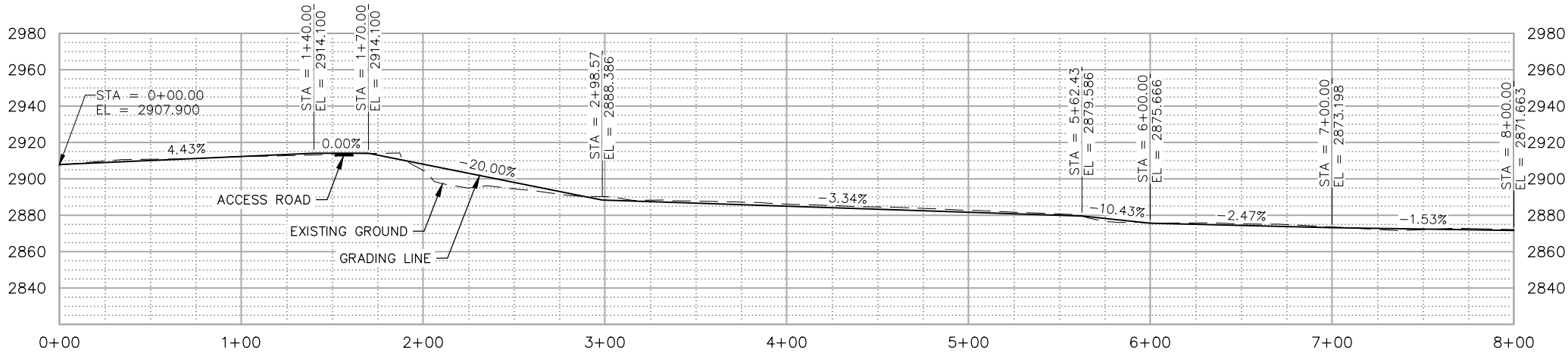
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SHEET 21 OF 26

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PLAN
SPILLWAY



PROFILE
SPILLWAY

SPILLWAY							
NUMBER	BEGIN NORTHING	BEGIN EASTING	LENGTH	RADIUS	LINE/CHORD DIRECTION	END NORTHING	END EASTING
L1	1275226.42	2195095.49	23.897		S12° 25' 34.92"E	1275203.08	2195100.63
C1	1275203.08	2195100.63	44.738	50.000	S13° 12' 23.45"W	1275160.97	2195090.75
L2	1275160.97	2195090.75	33.867		S38° 50' 21.82"W	1275134.59	2195069.51
C2	1275134.59	2195069.51	32.864	50.000	S20° 00' 35.53"W	1275104.26	2195058.47
L3	1275104.26	2195058.47	171.496		S01° 10' 49.23"W	1274932.80	2195054.94
C3	1274932.80	2195054.94	25.681	50.000	S13° 32' 02.64"E	1274908.10	2195060.88
L4	1274908.10	2195060.88	58.587		S28° 14' 54.51"E	1274856.50	2195088.61
C4	1274856.50	2195088.61	44.628	50.000	S53° 49' 05.59"E	1274831.02	2195123.45
L5	1274831.02	2195123.45	35.721		S79° 23' 16.67"E	1274824.44	2195158.56
C5	1274824.44	2195158.56	58.928	100.000	S62° 30' 23.22"E	1274797.63	2195210.07
L6	1274797.63	2195210.07	52.875		S45° 37' 29.77"E	1274760.65	2195247.87
C6	1274760.65	2195247.87	24.489	100.000	S52° 38' 25.95"E	1274745.82	2195267.29
L7	1274745.82	2195267.29	35.908		S59° 39' 22.12"E	1274727.68	2195298.27
C7	1274727.68	2195298.27	46.859	75.000	S41° 45' 26.46"E	1274693.29	2195328.98
C8	1274674.97	2195337.08	14.461	75.000	S18° 20' 04.83"E	1274661.26	2195341.62
L9	1274661.26	2195341.62	74.961		S12° 48' 38.85"E	1274588.16	2195358.25

SHEET NOTES:

- SPILLWAY EROSION PROTECTION MAT SHALL BE FLEXAMAT 10NW OR APPROVED EQUAL, SEE SPECIFICATION SECTION 02380. PLAN QUANTITY (16,000 SF) IS FACTORED TO INCLUDE TOE IN/ANCHOR TRENCH
- FG PROFILE DISPLAYED ON THIS SHEET IS FOR BASE BID. IT DOES NOT REPRESENT SUBGRADE FOR ADDITIVE ALTERNATIVE 1.
- SEE SHEET SW2 FOR EROSION PROTECTION MAT INSTALLATION DETAILS.



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SPILLWAY IMPROVEMENTS
PLAN AND PROFILE

BAR ISLAND DAM MODIFICATIONS

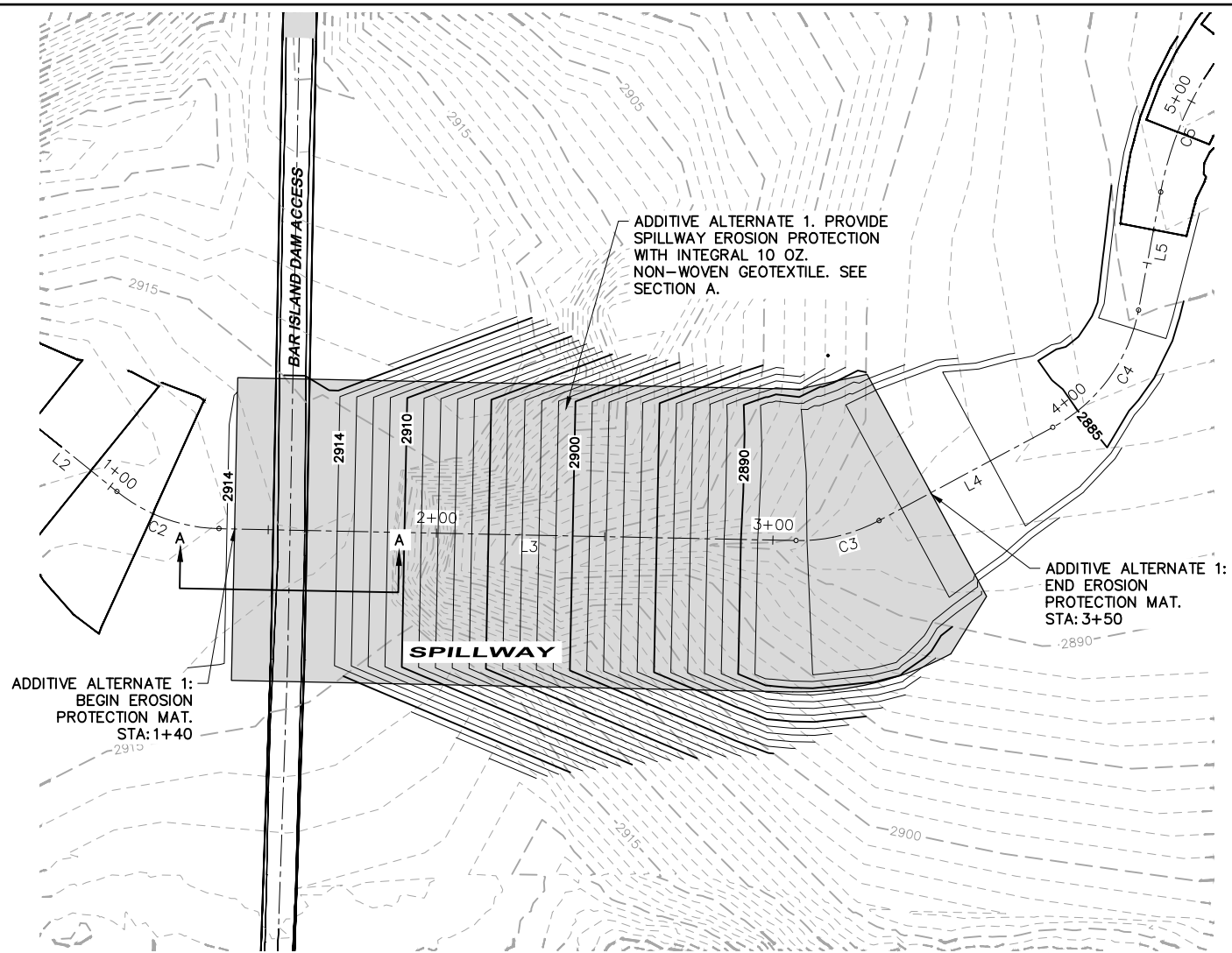
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DESIGNED BY: DAR				
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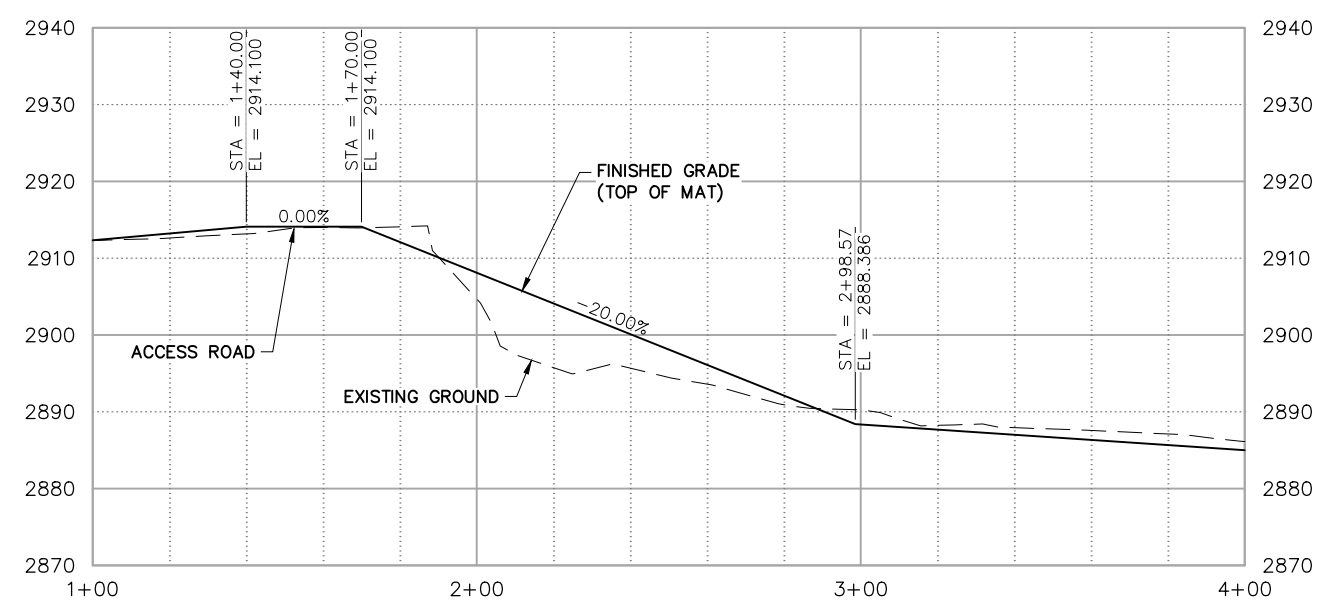
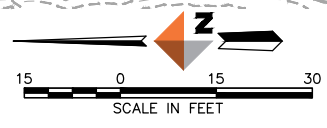
SW1

SHEET 22 OF 26

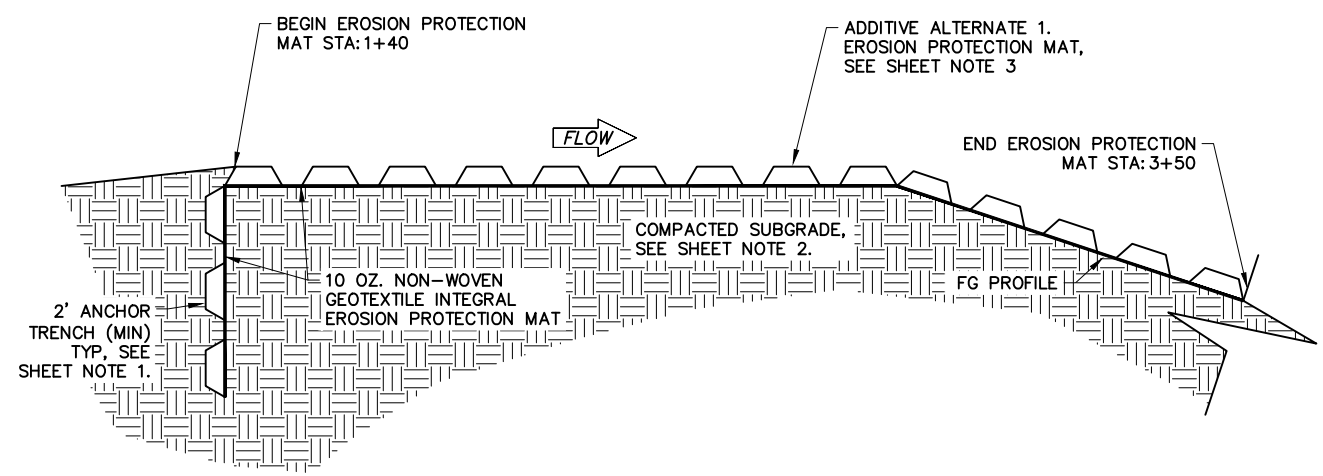
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PLAN
SPILLWAY



PROFILE
SPILLWAY



A SECTION
EROSION PROTECTION MAT

SHEET NOTES:

1. EROSION PROTECTION MAT SHALL BE ANCHORED IN A MINIMUM OF 2 FEET ON ALL SIDES, OR AS REQUIRED BY MANUFACTURER.
2. SUBGRADE SURFACE SHALL BE SMOOTH AND FREE OF ROCKS, ROOTS, DEBRIS, AND OTHER PROTRUSIONS THAT WOULD PREVENT INTIMATE CONTACT BETWEEN EROSION PROTECTION MAT AND THE SUBGRADE. ENSURE THE PREPARED SUBGRADE PROVIDES A SMOOTH, FIRM, AND UNYIELDING FOUNDATION FOR EROSION PROTECTION MAT.
3. INSTALL MAT PER MANUFACTURERS RECOMMENDATIONS.
4. CONSTRUCT THE EROSION PROTECTION MAT WITH AN ANCHOR TRENCH IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.



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UNITED STATES DEPARTMENT OF THE INTERIOR
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**SPILLWAY IMPROVEMENTS
PLAN, PROFILE, & DETAILS**

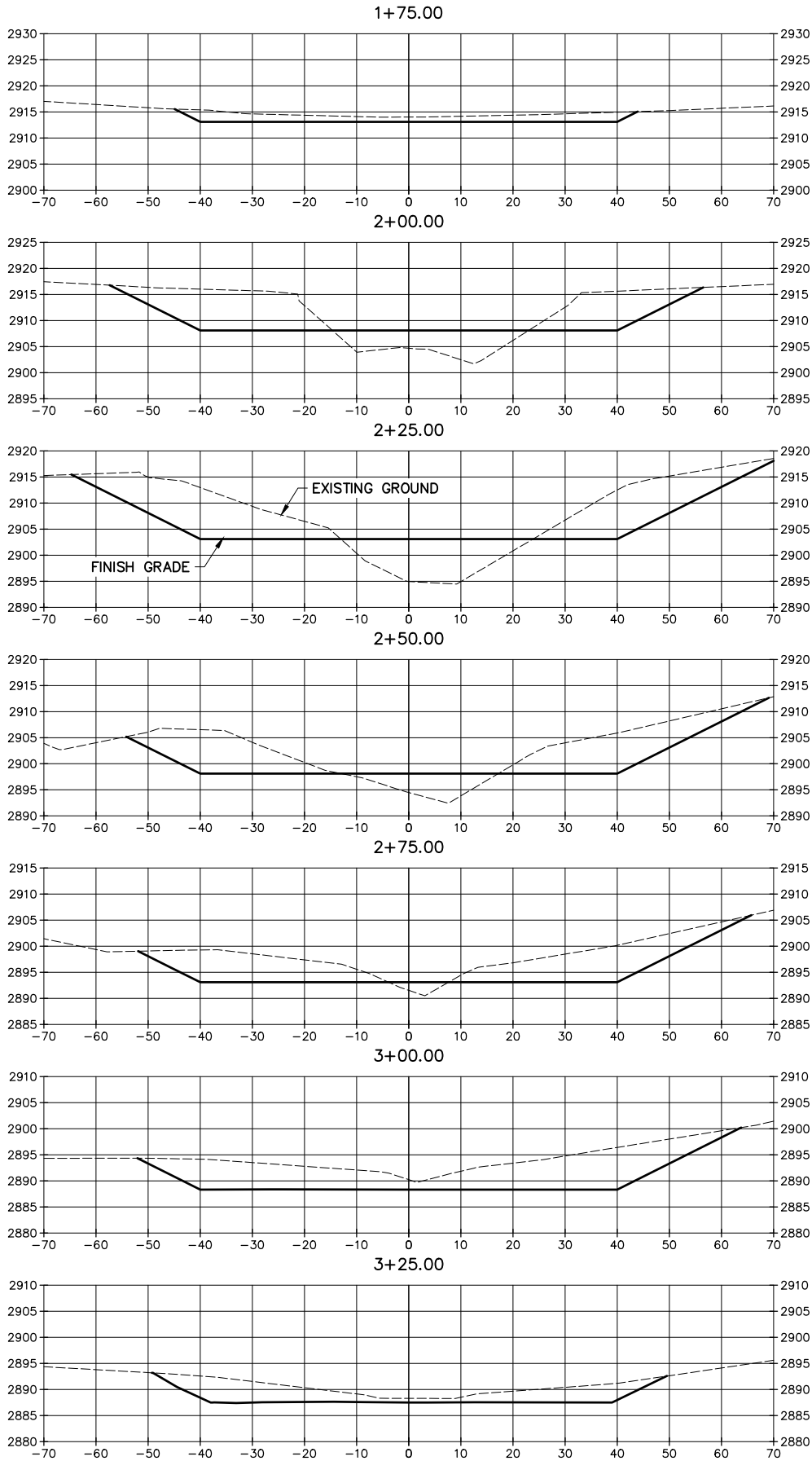
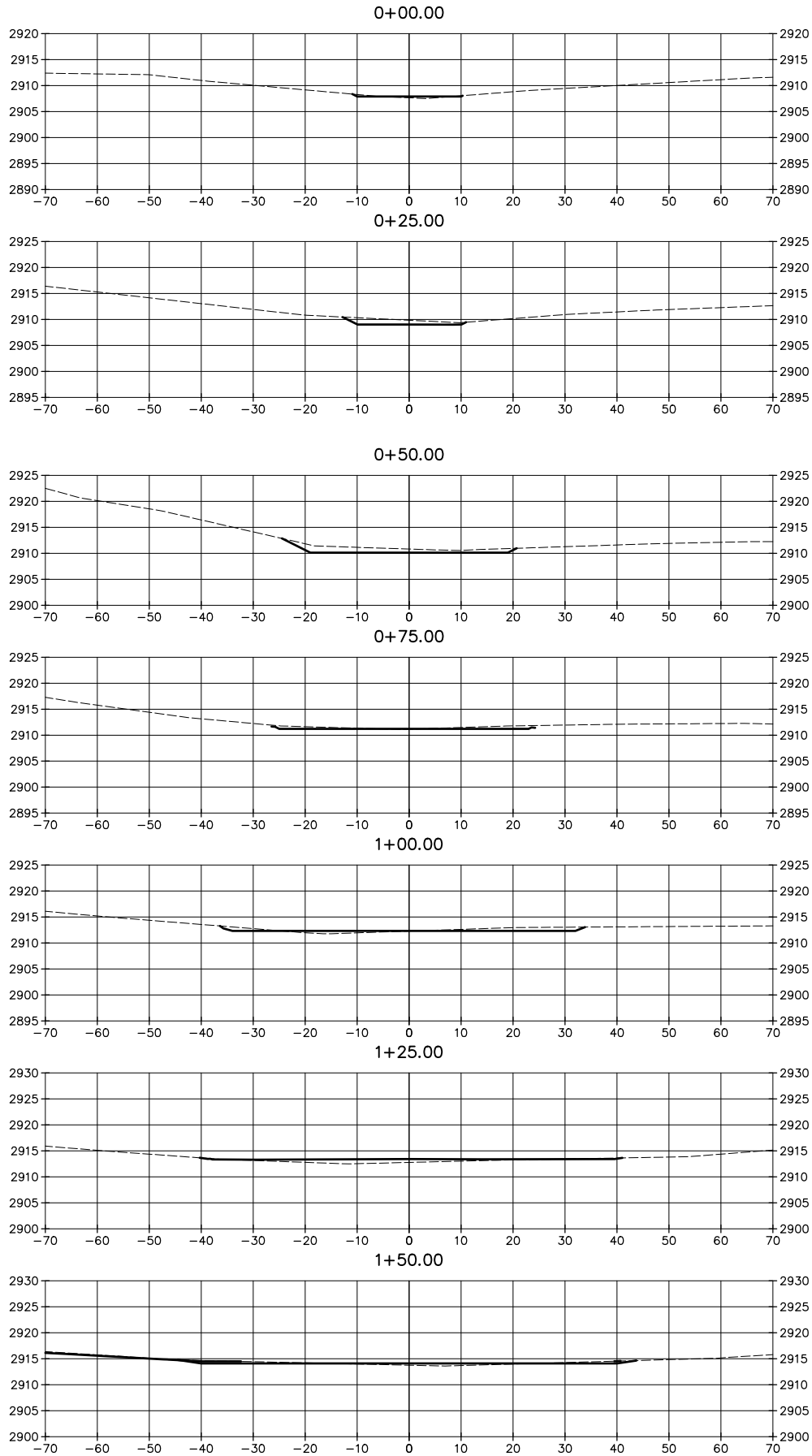
BAR ISLAND DAM MODIFICATIONS

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DESIGN OFFICE: DOWL (FIELD SURVEY BY JACKOLA)
DESIGNED BY: DAR DATE: 12-22-25
DRAWN BY: IME DATE: 12-22-25
CHECKED BY: RCR DATE: 12-22-25
APPROVED BY: DAR DATE: 12-22-25

SW2
SHEET 23 OF 26

MONTANA
MALTA

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CROSS SECTION NOTES:

1. FG SURFACE IS BASE BID.
2. EROSION PROTECTION MAT (ADD ALT 1) IS NOT SHOWN FOR CLARITY

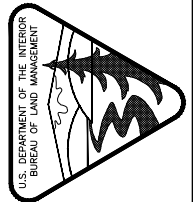


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SPILLWAY CROSS SECTION
STA. 0+00 TO STA. 3+25

BAR ISLAND DAM MODIFICATIONS

MONTANA

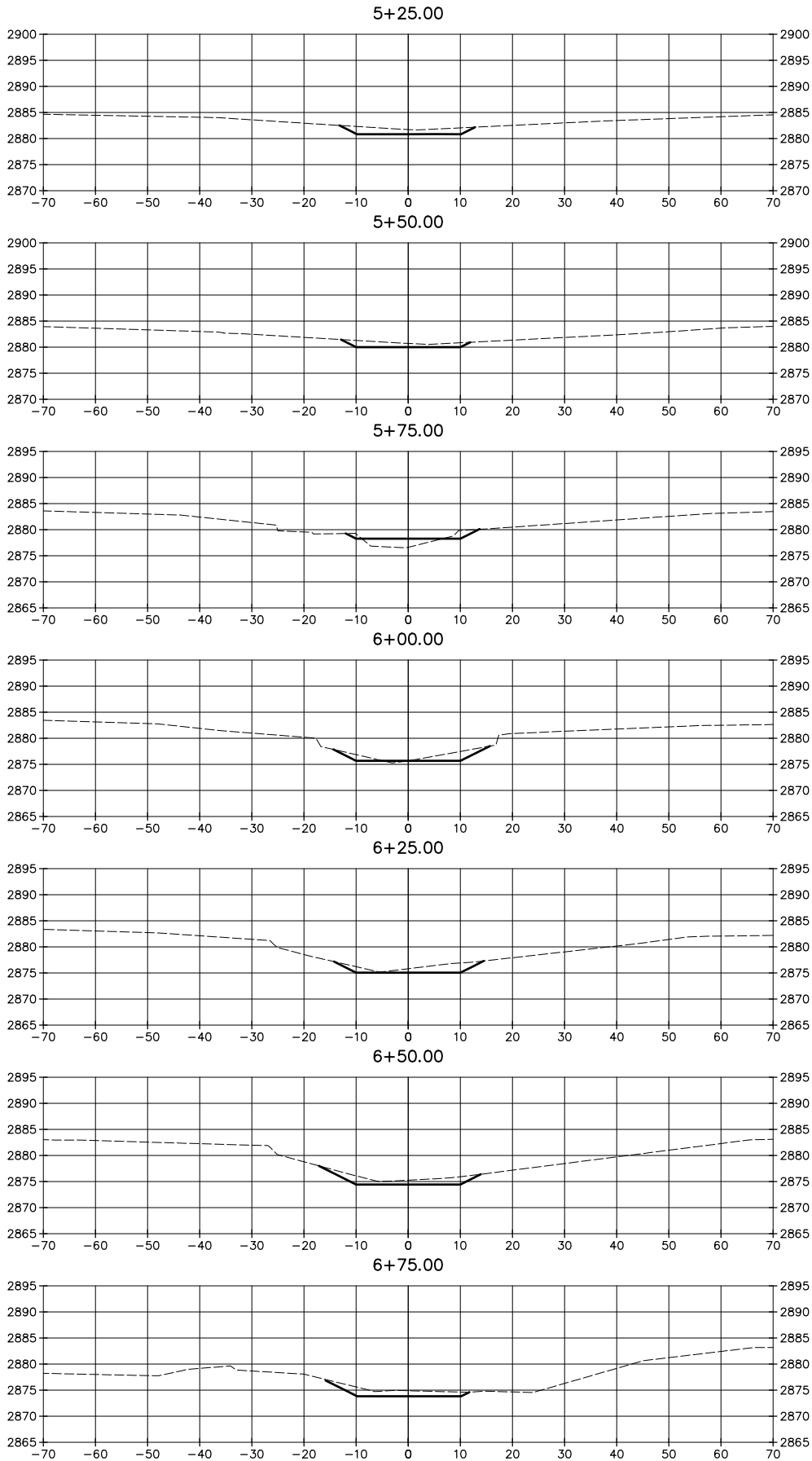
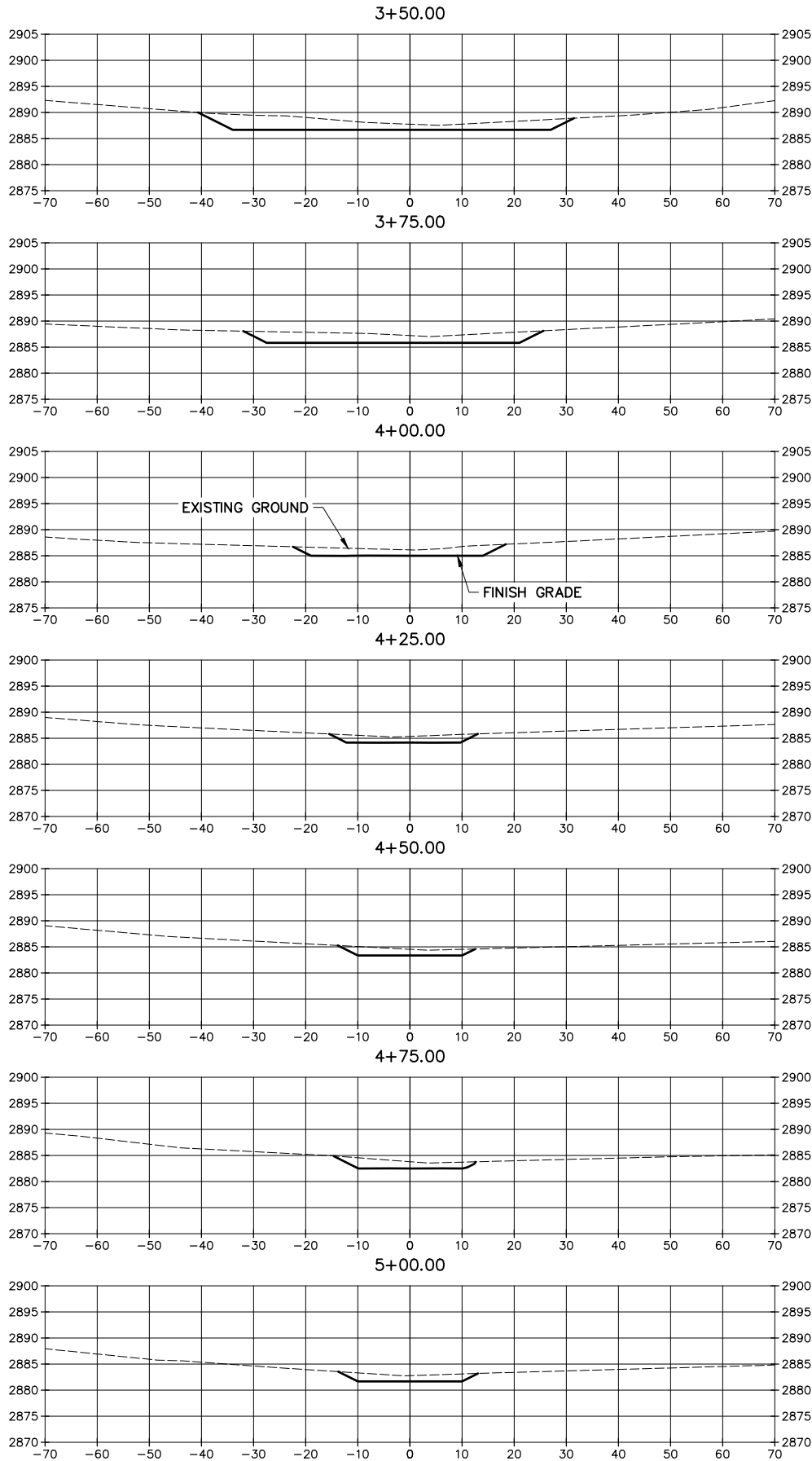
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AUTOCAD NAME:	SC-CS-ASW-12652-01.DWG	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY:	DAR				
DRAWN BY:	IME				
CHECKED BY:	RGR				
APPROVED BY:	DAR				

SWX1

SHEET 24 OF 26

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SPILLWAY CROSS SECTION
STA. 3+50 TO STA. 6+75

BAR ISLAND DAM MODIFICATIONS

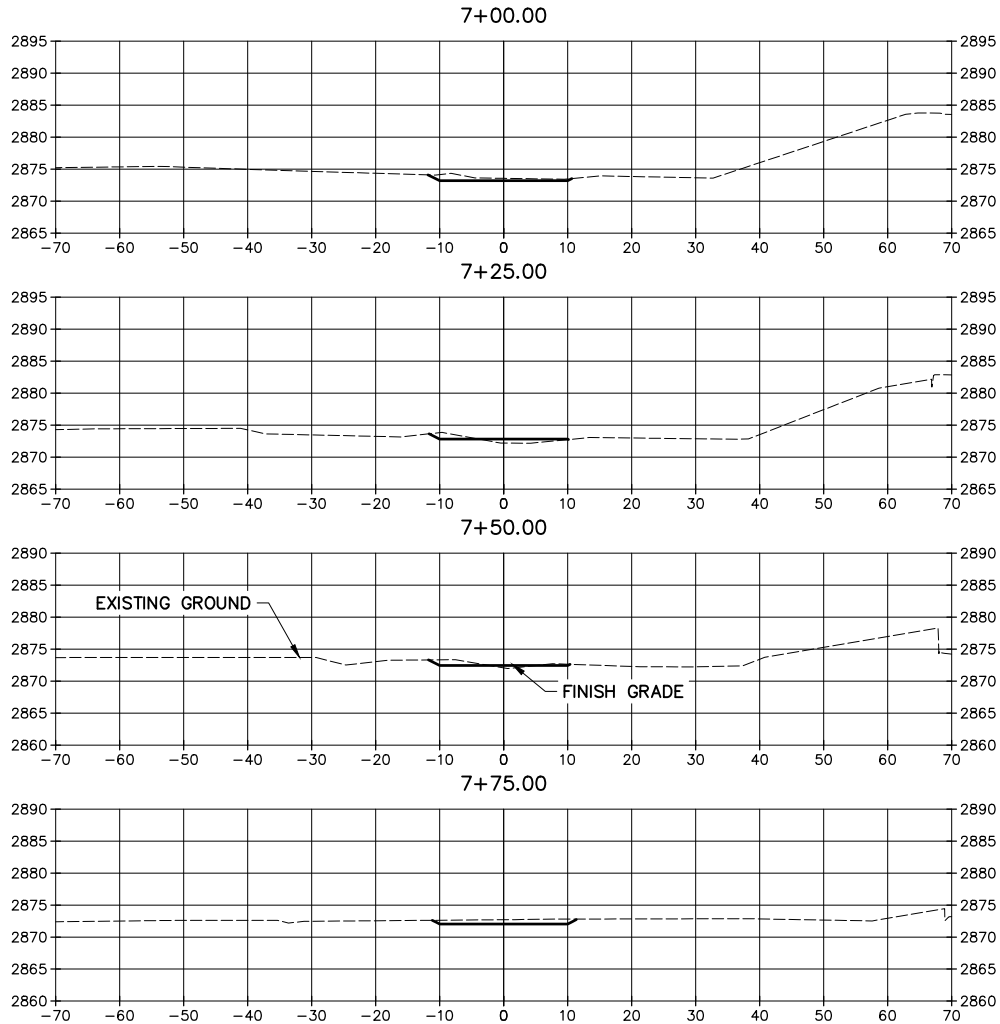
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AUTOCAD NAME:	SC-CS-ASW-12652-01.DWG	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY:	DAR			DATE: 12-22-25	
DRAWN BY:	IME			DATE: 12-22-25	
CHECKED BY:	RCR			DATE: 12-22-25	
APPROVED BY:	DAR			DATE: 12-22-25	

SWX2

SHEET 25 OF 26

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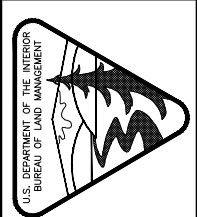




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STA. 7+00 TO STA. 7+75

BAR ISLAND DAM MODIFICATIONS

MONTANA

AUTOCAD NAME:	SC-CS-ASW-12652-01.DWG	MARK	REVISION	DATE	APPROVED
DESIGN OFFICE:	DOWL (FIELD SURVEY BY JACKOLA)				
DESIGNED BY:	DAR			DATE: 12-22-25	
DRAWN BY:	IME			DATE: 12-22-25	
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SWX3

SHEET 26 OF 26